Clinical Study Protocol

PHASE 2 STUDY OF LINVOSELTAMAB IN PATIENTS WITH SMOLDERING MULTIPLE MYELOMA AT HIGH RISK OF PROGRESSION TO MULTIPLE MYELOMA

Compound:	Linvoseltamab (REGN5458)
Clinical Phase:	Phase 2
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LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Ab+	Antibody positive
ADA	Anti-drug antibody
ADL	Activities of daily living
AE	Adverse event
AESI	Adverse event(s) of special interest
ALT	Alanine aminotransferase
AP	Appetite loss (EORTC QLQ-C30 symptom scale/item)
aPTT	Activated partial thromboplastin time
AST	Aspartate aminotransferase
ASTCT	American Society for Transplantation and Cellular Therapy
AxMP	Auxiliary medicinal product
BAFF	B cell activation factor
BCMA	B cell maturation antigen
β-hCG	β-human chorionic gonadotropin
BI	Body image (EORTC QLQ-MY20 scale)
BiPAP	Bilevel positive airway pressure
BLyS	B lymphocyte stimulator
BMPC	Bone marrow plasma cell
BUN	Blood urea nitrogen
C1D1	Cycle 1 day 1
C3D1	Cycle 3 day 1
CAR	Chimeric antigen receptor
CARTOX-10	CAR T cell therapy-associated toxicity – 10-point neurological assessment
CD3	Cluster of differentiation 3
CD25	Cluster of differentiation 25
CD38	Cluster of differentiation 38
CD69	Cluster of differentiation 69
C_{EOI}	Concentration at the end of infusion
CF	Cognitive functioning (EORTC QLQ-C30 functional scale)
CI	Confidence interval
C_{max}	Maximum (peak) concentration
CMV	Cytomegalovirus

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CNS	Central nervous system
CO	Constipation (EORTC QLQ-C30 symptom scale/item)
COVID-19	Coronavirus disease 19
CPAP	Continuous positive airway pressure
CR	Complete response
CRAB	Hyper**C**alcemia, **R**enal insufficiency, **A**nemia, or lytic **B**one lesions
CRF	Case report form (electronic or paper)
CRO	Contract research organization
CRP	C-reactive protein
CRS	Cytokine release syndrome
CSR	Clinical study report
CT	Computed tomography
CTC	Circulating tumor cell
CTCAE	Common Terminology Criteria for Adverse Events
CTFG	Clinical Trial Facilitation Group
C_{trough}	Lowest concentration in a dosing interval
CYP450	Cytochrome P450
DI	Diarrhea (EORTC QLQ-C30 symptom scale/item)
DNA	Deoxyribonucleic acid
DOR	Duration of response
DS	Disease symptoms (EORTC QLQl-MY20 scale)
DY	Dyspnea (EORTC QLQ-C30 symptom scale/item)
EBV	Epstein-Barr virus
EC	Ethics Committee
ECG	Electrocardiogram
Echo	Echocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	Electronic case report form
EDC	Electronic data capture
EF	Emotional functioning (EORTC QLQ-C30 functional scale)
EMP	Extramedullary plasmacytoma
EOI	End of infusion
EORTC QLQ-C30	European Organization for Research and Treatment of Cancer Quality of Life Questionnaire-Core 30
EORTC QLQ-MY20	EORTC Quality of Life Questionnaire -Multiple Myeloma Module 20

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EU	European Union
FA	Fatigue (EORTC QLQ-C30 symptom scale/item)
FACIT Item GP5	Functional Assessment of Chronic Illness Therapy Item GP5
FAS	Full analysis set
FBR	Future biomedical research
FDA	Food and Drug Administration
FDG	F-fluorodeoxyglucose
FDG-PET/CT	F-fluorodeoxyglucose positron emissions tomography/computed tomography
FI	Financial impact (EORTC QLQ-C30 scale)
FIH	First-in-human
FISH	Fluorescent in situ hybridization
FLC	Free light chain
FP	Future perspective (EORTC QLQ-MY20 scale)
FSH	Follicle stimulating hormone
GCP	Good Clinical Practice
G-CSF	Granulocyte colony stimulating factor
GFR	Glomerular filtration rate
GHS	Global health status (EORTC QLQ-C30 scale)
HBV	Hepatitis B virus
HCV	Hepatitis C virus
HBsAg	Hepatitis B surface antigen
HHV-6	Human herpesvirus 6
HIV	Human immunodeficiency virus
HRQoL	Health-related quality of life
IA	Interim analysis
IB	Investigator's Brochure
ICANS	Immune effector cell-associated neurotoxicity syndrome
ICE	Immune effector cell-associated encephalopathy
ICF	Informed consent form
ICH	International Council for Harmonisation
IDSA	Infectious Diseases Society of America
IEC	Institutional Ethics Committee
IFN	Interferon
IgA	Immunoglobulin A

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IgD	Immunoglobulin D
IgE	Immunoglobulin E
IgG	Immunoglobulin G
IgM	Immunoglobulin M
IHC	Immunohistochemistry
IL-2	Interleukin-2
IL-6	Interleukin-6
IL-10	Interleukin-10
IMP	Investigational medicinal product
IMWG	International Myeloma Working Group
INR	International normalized ratio
IRR	Infusion-related reaction
IUD	Intrauterine device
IUS	Intrauterine hormone-releasing system
IV	Intravenous
LAM	Lactational amenorrhea method
LDH	Lactate dehydrogenase
MDRD	Modification of Diet in Renal Disease
MedDRA	Medical Dictionary for Regulatory Activities
MGUS	Monoclonal gammopathy of undetermined significance
MM	Multiple myeloma
MRD	Minimal residual disease
MRI	Magnetic resonance imaging
MS	Mass spectrometry
MTD	Maximum tolerated dose
MUGA	Multi-gated acquisition scan
NCCN	National Comprehensive Cancer Network
NCI-CTCAE	National Cancer Institute-Common Terminology Criteria for Adverse Events
NGS	Next-generation sequencing
NK	Natural killer
NV	Nausea and vomiting (EORTC QLQ-C30 symptom scale/item)
NYHA	New York Heart Association
OCs	Operating characteristics
ORR	Overall response rate

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OS	Overall survival
PA	Pain (EORTC QLQ-C30 symptom scale/item)
PCR	Polymerase chain reaction
PCSV	Potentially clinically significant value
PD	Progressive disease
PET/CT	Positron emissions tomography/computed tomography
PETHEMA	Programa de Estudio y Tratamiento de las Hemopatías Malignas (The Spanish Society of Hematology)
PF	Physical functioning (EORTC-C30 functional scale)
PFS	Progression-free survival
PFS2	PFS after initiation of treatment for MM
PGIC	Patient Global Impression of Change
PGIS	Patient Global Impression of Severity
PJP	*Pneumocystis jirovecii* pneumonia
PK	Pharmacokinetic
PO	Oral
PR	Partial response
PRO	Patient-reported outcome
PT	Prothrombin time
PTT	Partial thromboplastin time
QW	Weekly
Q2W	Every 2 weeks
Q4W	Every 4 weeks
Q12W	Every 12 weeks
Q24W	Every 24 weeks
QLQ-C30	Quality of Life Questionnaire-Core 30
QLQ-MY20	Quality of Life Questionnaire-Multiple Myeloma Module 20
QoL	Quality of life
QTcF	Heart rate-corrected QT interval (Fridericia's correction formula)
RANKL	Receptor activator of nuclear factor kappa-B ligand
RBC	Red blood cell
RBQM	Risk-Based Quality Management
Regeneron	Regeneron Pharmaceuticals, Inc.
RF	Role functioning (EORTC QLQ-C30 functional scale)
RNA	Ribonucleic acid

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RRMM	Relapsed/refractory multiple myeloma
SAE	Serious adverse event
SAF	Safety analysis set
SAP	Statistical analysis plan
SAS	Statistical Analysis System
sBCMA	Soluble BCMA
S_{Cr}	Serum creatinine
SDR	Source data review
SDV	Source data verification
SE	Side effects of treatment (EORTC QLQ-MY20 scale)
SEQ	Sequencing
SF	Social functioning (EORTC QLQ-C30 functional scale)
SI	International System of Units
SIFE	Serum immunofixation
SL	Insomnia (EORTC QLQ-C30 symptom scale/item)
SLAMF7	Signaling lymphocytic activation molecule family member 7
SLiM	Greater than or equal to **S**ixty percent clonal plasma cells in the bone marrow, involved/uninvolved free **Li**ght chain ratio of ≥100 with the involved FLC being ≥100 mg/L, **M**RI with >1 focal lesion
SMM	Smoldering multiple myeloma
SMQ	Standardized MedDRA query
SOC	System organ class
SPEP	Serum protein electrophoresis
SSET	Study Safety Evaluation Team
SUSAR	Suspected unexpected serious adverse reaction
TEAE	Treatment-emergent adverse event
trSAE	Treatment-related serious adverse event
TLS	Tumor lysis syndrome
TNF	Tumor necrosis factor
TTR	Time to response
UIFE	Urine immunofixation
ULN	Upper limit of normal
UPEP	Urine protein electrophoresis
US	United States
VGPR	Very good partial response

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W1D1	Week 1 day 1
WBC	White blood cell
WES	Whole exome sequencing
WGS	Whole genome sequencing
WOCBP	Women of child-bearing potential

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CLINICAL STUDY PROTOCOL SYNOPSIS

Title	Phase 2 Study of Linvoseltamab in Patients with Smoldering Multiple Myeloma at High Risk of Progression to Multiple Myeloma
Short Title	Linvoseltamab for High-Risk Smoldering Multiple Myeloma
Site Location(s)	Approximately 15 sites in Spain
Objective(s)	
Primary:	
Safety Run-In (Part 1)	To evaluate the safety and tolerability of linvoseltamab in participants with high-risk smoldering multiple myeloma (SMM)
Expansion (Part 2)	To evaluate the preliminary clinical activity of linvoseltamab by assessment of complete response (CR) and minimal residual disease (MRD) negativity in participants with high-risk SMM, including those from the safety run-in cohort
Secondary:	The secondary objectives of the study are (applicable to both safety run-in [Part 1] and expansion [Part 2] unless otherwise specified): • To characterize the safety of linvoseltamab (Part 2) • To evaluate the overall response rate (ORR) • To evaluate the duration of response (DOR) • To evaluate biochemical progression-free survival (PFS) • To evaluate the proportion of participants that achieve MRD negativity • To evaluate the proportion of participants with sustained MRD negativity • To evaluate the time to progression to multiple myeloma (MM) • To evaluate clinical PFS • To measure time to the start of systemic therapy for MM • To evaluate overall survival (OS) • To characterize the pharmacokinetic (PK) properties of linvoseltamab • To assess the immunogenicity of linvoseltamab

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Study Design	This is an open-label, phase 2 study designed to evaluate the safety and efficacy of linvoseltamab in participants with SMM who are at a high risk of progression to MM. As this is the first time that linvoseltamab will be evaluated in this population, enrollment will begin with a safety run-in part to assess the suitability of the recommended dosing regimen. After early safety and tolerability are established, an expansion part will open to confirm the safety and evaluate the preliminary clinical activity of single-agent linvoseltamab. **Safety Run-In (Part 1):** At least 6 participants will receive linvoseltamab monotherapy starting with a step-up schedule that consists of weekly dose increments of 1 mg (week 1), 4 mg (week 2), and 25 mg (week 3) before reaching a full dose of 200 mg. Each participant must be monitored in an inpatient setting for at least 24 hours after each step-up dose infusion as a precautionary safety measure. The step-up period is expected to last 3 weeks unless a tolerability issue requires repeat administration of step-up doses. All subsequent treatment will be given over 28-day cycles. Full-dose linvoseltamab will be administered weekly (QW) for the first cycle and then every 2 weeks (Q2W) for the next 4 cycles. At the start of cycle 6, the dosing frequency will decrease to every 4 weeks (Q4W) and continue until completion of 24 cycles or fulfillment of a protocol-defined criterion for study drug discontinuation. **Expansion (Part 2):** An additional 34 participants will be enrolled to ensure at least 40 participants receive linvoseltamab monotherapy according to the same dosing schedule established in the safety run-in part. The expansion cohort will open after careful review of all the safety and efficacy data gathered from the 6 participants treated in the safety run-in cohort.
Study Duration	For each participant, the study will span the following 3 periods: **Screening Period:** Screening procedures may occur over a period of up to 28 days for each study participant, ending on the day preceding the planned first dose of linvoseltamab. **Treatment Period:** The treatment period starts on the day when the first dose of linvoseltamab is administered and continues until completion of the end of treatment visit or, if the end of treatment visit does not occur, 30 days after the last dose of linvoseltamab is administered. Treatment may continue for up to 24 cycles or until fulfillment of a protocol-defined criterion for study drug discontinuation. **Follow-Up Period (up to 5 years after last dose):** • **Post-treatment follow-up:** Post-treatment follow-up starts after the end of treatment visit and involves ongoing collection of safety and efficacy data. Safety monitoring will continue for up to 90 days after the last dose of linvoseltamab. Participants with an ongoing biochemical response will continue to undergo routine efficacy

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assessments until confirmed biochemical progression, start of non-protocol cancer therapy, or death.

- **Long-term follow-up:** Long-term follow-up begins after post-treatment follow-up to collect survival status as well as other relevant long-term outcomes (eg, progression to MM, subsequent non-protocol cancer therapy, and stem cell collection) and will continue for up to 5 years after last dose of study drug, unless death or study termination by the sponsor occurs earlier. Participants may be followed remotely for survival and subsequent treatment history.

End of Study Definition

An individual participant will reach the end of study whenever one of the following events occurs: completion of 5 years of follow-up after the last dose of study drug, death, loss to follow-up, or withdrawal of consent.

The study/trial will end on the date of the last participant's last visit for any of the reasons listed above or study termination by the sponsor, whichever occurs earlier.

Population

Sample Size: At least 40 participants

Target Population: Patients diagnosed with SMM within 5 years of study enrollment per International Myeloma Working Group (IMWG) criteria defined as:

- Serum M-protein ≥30 g/L **or** urinary M-protein ≥500 mg per 24 hour **or** bone marrow plasma cells (BMPCs) 10% to <60% **AND**
- Absence of myeloma-defining events or other related conditions (SLiM CRAB*)

*SLiM (greater than or equal to **S**ixty percent clonal plasma cells in the bone marrow, involved/uninvolved free **Li**ght chain ratio of ≥100 with the involved free light chain (FLC) being ≥100 mg/L, **M**RI with >1 focal lesion) CRAB (hyper**C**alcemia, **R**enal insufficiency, **A**nemia, or lytic **B**one lesions)

Patients with high-risk SMM according to the following markers at screening:

- At least 2 of the following abnormalities:
  - Serum M-protein >20 g/L or 2 g/dL
  - Serum involved/uninvolved FLC ratio >20
  - BMPCs >20%

AND/OR

- Both of the below findings (The Spanish Society of Hematology / Programa de Estudio y Tratamiento de las Hemopatíás Malignas [PETHEMA] criteria):

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- ≥95% aberrant/clonal BMPCs
- Immunoparesis defined as reduction of ≥1 uninvolved immunoglobulin (Ig) isotype of at least 25% below the lower normal limit (only IgG, IgA and IgM will be considered)

Treatment(s)

Study Drug	Linvoseltamab
Dose/Route/Schedule:	1 mg (week 1), 4 mg (week 2), 25 mg (week 3), 200 mg (weeks 4+)

Endpoint(s)

Primary:	
Safety Run-In (Part 1)	Frequency of adverse events of special interest (AESI) including grade 2 or higher cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS) during the safety run-in observation period
	Frequency and severity of treatment-emergent adverse events (TEAEs) assessed during the safety run-in observation period
Expansion (Part 2)	Achievement of complete response (CR) as determined by the investigator using IMWG response criteria
	MRD negativity (10^{-5} sensitivity) at 12 and 24 months after the start of treatment
Secondary:	Frequency and severity of TEAEs/serious adverse events (SAEs) and laboratory abnormalities
	Overall response of partial response (PR) or better as measured using IMWG criteria
	DOR as measured using IMWG criteria
	Biochemical PFS as measured using IMWG criteria
	MRD negativity at any time among participants that achieve a response of VGPR or better
	Sustained MRD negativity assessed up to 3 years after end of treatment
	Time from treatment initiation to date of any myeloma-defining event per SLiM-CRAB diagnostic criteria
	Time from start of treatment to date of progression to MM or death, whichever happens first
	Time to initiation of first-line treatment for MM
	OS as measured from study entry until up to 5 years after end of treatment

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	Concentration of linvoseltamab in serum over time for the duration of treatment Incidence and titer of anti-drug antibodies (ADAs) to linvoseltamab over time for the duration of treatment
Procedures and Assessments	Screening: Medical/oncology history, demographics, complete physical examination, height, β2-microglobulin, echocardiogram or multi-gated acquisition scan (cardiac ejection fraction), viral serologies (human immunodeficiency virus [HIV], hepatitis B virus [HBV], hepatitis C virus [HCV]), coagulation parameters: prothrombin time (PT) / international normalized ratio (INR) and activated partial thromboplastin time (aPTT) / partial thromboplastin time (PTT), and whole-body imaging. Safety: Vital signs, weight, brief neurologic examination, electrocardiogram, Eastern Cooperative Oncology Group (ECOG) performance status, laboratory assessments, adverse events (AEs), and concomitant medications. Efficacy: Serum protein electrophoresis and immunofixation (SPEP/SIFE), 24-hour urine collection for urine protein electrophoresis and immunofixation (UPEP/UIFE), serum FLC testing, immunoglobulin quantification (IgA, IgM, IgG, IgD, IgE), and bone marrow aspirate/biopsy. Additional biomarker assessments and patient-reported outcome measurements. Blood samples for drug concentration analysis and ADA assessment in serum will be collected.
Statistical Plan	The statistical plan may be revised during the study to accommodate amendments to the clinical study protocol if necessary.
Statistical Hypothesis:	This is an exploratory study to evaluate the safety and preliminary clinical activity of linvoseltamab, and therefore no formal statistical hypothesis testing is planned.
Justification of Sample Size:	At least 6 participants will be enrolled during the safety run-in part of the study and 34 participants will be enrolled in the expansion part with the selected dosing regimen for expansion. These will be combined to reach a minimum total sample size of 40 participants treated with the selected dosing regimen and who have at least one post-baseline response assessment. With this sample size, we can estimate the ≥CR rate with reasonable precision and explore preliminary clinical activity, safety, PK properties, and biomarker changes associated with the dosing regimen.
Efficacy Analyses	The full analysis set (FAS) will include all participants who received at least 1 dose (any amount) of study drug and had at least one post-baseline response assessment. Efficacy endpoints will be analyzed using the FAS.

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Primary efficacy analyses: The primary endpoints are the incidence of CR or better and MRD negativity determined by IMWG response criteria.

The rate of MRD negativity is defined as the proportion of participants who achieve MRD negativity at 10^{-5}, among participants with very good partial response or better (≥VGPR) based on FAS. MRD-negative rate at 12 months and at 24 months after treatment initiation will be summarized. For the primary analysis, participants with missing data will be treated as non-responders and will be counted in the denominator, but not in the numerator. The point estimate and exact 95% confidence interval (CI) will be provided using the Clopper-Pearson method.

The ≥CR rate is defined as the proportion of participants with a best overall response of CR or better by IMWG response criteria. For the primary analysis, participants with inevaluable response will be considered as non-responders. The point estimate and exact 95% CI will be provided using the Clopper-Pearson method.

Secondary efficacy analyses: ORR is defined as the proportion of treated participants who achieve a best response of PR or better. The ORR and the corresponding 95% CI will be calculated using the Clopper-Pearson method.

DOR is defined for responders (participants who achieve PR or better) only. It is the time from the date of the first documented response (at least PR) until the date of first documented disease progression or death due to any cause, whichever occurs first. For participants who are alive and have not experienced disease progression at the time of analysis, DOR will be censored at their latest available adequate disease assessments.

PFS is defined as the time from first dose date until the first date of progressive disease, or death due to any cause, whichever occurs first. For participants who are alive and have not experienced disease progression at the time of analysis, PFS will be censored at their latest available adequate disease assessments.

Time to new anti-myeloma therapy (ie, new systemic therapy for MM) is measured as the time from the start of study treatment to the initiation of systemic therapy for MM, or death due to any cause, whichever occurs first. If a participant is not known to have died or have received new systemic therapy for MM at the time of analysis, this endpoint will be censored at the last date the participant is documented to be receiving study treatment or to be free of new anti-myeloma therapy.

OS is measured as the time from the start of study treatment until death due to any cause. If a participant is not known to have died at the time of analysis, OS will be censored at the last date that the participant is documented to be alive.

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	The Kaplan-Meier curve and estimated median (or an appropriate percentile if the median is not reached) along with its 95% CI will be provided for each of the secondary time-to-event endpoints described above. Time to response (TTR) is defined for responders (participants that achieve PR or better) only. It is the time from first dose date to the date of first response. Descriptive statistics will be presented for TTR.
Safety Analyses	All AEs reported in this study will be coded using the Medical Dictionary for Regulatory Activities (MedDRA®). Summary of TEAEs, deaths and other SAEs, and TEAEs leading to permanent treatment discontinuation will be summarized. Other safety, including vital signs (temperature, pulse, blood pressure, pulse oximetry [on dosing days], and respiration rate) and laboratory tests will be summarized by baseline and change from baseline to each scheduled assessment time point with descriptive statistics.
Interim Analysis	Interim analyses for futility will be performed after 15 and 30 participants have started treatment with linvoseltamab and completed at least 15 weeks of follow-up with at least one post-baseline response assessment. The futility criteria for the first interim analysis are observing 5 or fewer responders or no CR. The futility criteria for the second interim analysis are observing 13 or fewer responders or 2 or fewer CRs. Enrollment will not be paused during interim analyses.

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1. INTRODUCTION

1.1. Overview of Smoldering Multiple Myeloma

Smoldering multiple myeloma (SMM) is a clonal plasma cell disorder that evolves from MGUS and represents a more advanced precursor stage before progression to MM. Diagnosis usually occurs incidentally because the condition is asymptomatic, although a recent population-based screening study estimated a prevalence of 0.5% in individuals older than 40 years of age (Rögnvaldsson, 2021). The rate of progression to MM is variable, reflecting the biological heterogeneity of this clinically defined entity. Prognostic factors associated with progression can help identify high-risk SMM, a subgroup of patients with an average time to progression to MM of 24 months (~50% risk of progression to MM at 2 years). Most of these patients are considered to have early malignancy but have yet to develop clinical manifestations of MM such as hypercalcemia, renal insufficiency, anemia, or lytic bone lesions (CRAB features) (Table 2).

Different risk models have been developed to refine the identification of truly high-risk SMM patients. The IMWG endorsed the "2-20-20" model based on serum M protein (>2 g/dL), FLC ratio (>20), and BMPCs (>20%), designating high-risk patients as those meeting at least 2 of the 3 factors (Lakshman, 2018) (Mateos, 2020). Alternatively, the Spanish model requires the presence of both ≥95% clonal plasma cells in the bone marrow by flow cytometry and immunoparesis (Pérez-Persona, 2007) whereas the evolving pattern considers the rate of serum M-protein rise defined as an increase of serum M protein ≥10% in the last 12 months (Fernández de Larrea, 2018). Additional models incorporating imaging and/or molecular features are under development.

1.2. Unmet Medical Need in High-Risk Smoldering Multiple Myeloma

Patients diagnosed with SMM have traditionally undergone observation until progression to MM; however, waiting until end-organ damage can introduce significant irreversible morbidity that is detrimental to patient outcomes. This paradigm is changing as risk stratification methods improve and better treatments emerge. Early intervention is now considered appropriate for SMM patients whose risk of progression is 50% or higher within 2 years. Two phase 3 clinical trials have demonstrated how early treatment in high-risk SMM with lenalidomide plus dexamethasone or lenalidomide alone can confer significant benefit over observation in terms of delaying progression to MM with end-organ damage (similar hazard ratios of ~0.2) (Mateos, 2013) (Lonial, 2020). One of the studies also demonstrated an overall survival benefit (Mateos, 2022). These improved outcomes were observed despite infrequent deep responses (VGPR or CR), suggesting that clonal control can be achieved without major responses. Nonetheless, many ongoing trials in the high-risk SMM space are evaluating more intensive combination therapy including consolidation with autologous stem cell transplantation as potentially curative approaches. While aggressive treatment strategies that mirror MM management are anticipated to induce deeper responses, not all patients are candidates for such treatment and the net benefit may be offset by increased toxicity, clonal selection, and altered immune regulation. Thus, exploring novel immunotherapy strategies that enhance immune clearance of clonal plasma cells with reduced side effects is highly relevant.

Additional background information on the study drug and development program can be found in the linvoseltamab IB.

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2. STUDY OBJECTIVES AND ENDPOINTS

The primary, secondary, and exploratory objectives and endpoints of the study are shown in Table 1.

Table 1: Study Objectives and Endpoints

Objective	Endpoint
Primary	
Safety Run-In (Part 1): To evaluate the safety and tolerability of linvoseltamab in participants with high-risk SMM	Frequency of AESI including grade 2 or higher CRS and ICANS during the safety run-in observation period Frequency and severity of TEAEs assessed during the safety run-in observation period
Expansion (Part 2): To evaluate the preliminary clinical activity of linvoseltamab by assessment of CR and MRD negativity in participants with high-risk SMM, including those from the safety run-in cohort	Achievement of CR as determined by the investigator using IMWG response criteria MRD negativity (10^{-5} sensitivity) at 12 and 24 months after the start of treatment
Secondary *Applicable to both safety run-in and expansion unless otherwise specified*	
To characterize the safety of linvoseltamab (Part 2)	Frequency and severity of TEAEs/SAEs and laboratory abnormalities
To evaluate the ORR	Overall response of PR or better as measured using IMWG criteria
To evaluate the DOR	DOR as measured using IMWG criteria
To evaluate biochemical PFS	Biochemical PFS as measured using IMWG criteria
To evaluate the proportion of participants that achieve MRD negativity	MRD negativity at any time among participants that achieve a response of VGPR or better
To evaluate the proportion of participants with sustained MRD negativity	Sustained MRD negativity for up to 3 years after end of treatment
To evaluate the time to progression to MM	Time from treatment initiation to date of any myeloma-defining event per SLiM CRAB diagnostic criteria (Table 2)
To evaluate clinical PFS	Time from start of treatment to date of progression to MM or death, whichever happens first
To measure time to the start of systemic therapy for MM	Time to initiation of first-line treatment for MM
To evaluate OS	OS as measured from study entry until up to 5 years after end of treatment
To characterize the PK properties of linvoseltamab	Concentration of linvoseltamab in serum over time for the duration of treatment
To assess the immunogenicity of linvoseltamab	Incidence and titer of ADAs to linvoseltamab over time for the duration of treatment

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Objective	Endpoint
Exploratory	
To evaluate PFS after initiation of treatment for MM (PFS2)	PFS from start of therapy for MM to date of progression or death, whichever happens first
To assess the ability to collect stem cells after linvoseltamab administration	Collection yield of CD34+ stem cells after treatment with linvoseltamab
To evaluate changes in soluble BCMA levels in serum	Concentration of total soluble BCMA in serum during treatment with linvoseltamab
To detect serum M-protein by MS and correlate with clinical response characteristics	Serum M-protein measurement by MS over time
To evaluate CTCs	CTC enumeration and phenotypic characterization over time
To characterize the effects of linvoseltamab on various immune cells	Numeric and phenotypic changes in B, T, NK, and monocyte cells in peripheral blood and bone marrow during treatment with linvoseltamab
To evaluate the genomic determinants of SMM and progression to MM	Whole exome DNA and RNA sequencing of neoplastic plasma cells and bone marrow immune cells
To evaluate the effects of linvoseltamab on polyclonal immunoglobulins	Concentrations of IgA, IgM, IgG, IgD, and IgE over time
To evaluate plasma cell surface antigens	Expression of BCMA and other surface antigens on plasma cells from bone marrow biopsies
To evaluate the effects of linvoseltamab on patient-reported health-related quality of life, functioning, and symptoms	Change from baseline in patient-reported symptoms, functioning, and global health status/quality of life per EORTC QLQ-C30 and EORTC QLQ-MY20 (2 global anchors PGIS and PGIC will be used to aid PRO interpretation)
To evaluate treatment effects of linvoseltamab on patient-reported overall impact of treatment toxicity as measured by the FACIT Item GP5	Change over time in patient-reported overall impact of treatment toxicity per the FACIT Item GP5
To evaluate healthcare resource use attributed to hospitalizations in participants receiving linvoseltamab	Number and duration of hospitalizations (including general ward, intensive care unit, emergency room visits) from start of study treatment to 90 days after last dose of study treatment

ADA=anti-drug antibody; AE=adverse event; AESI=adverse event(s) of special interest; BCMA=B cell maturation antigen; CR=complete response; CRAB= hyperCalcemia, Renal insufficiency, Anemia, or lytic Bone lesions; CRS=cytokine release syndrome; DNA=deoxyribonucleic acid; DOR=duration of response; EORTC QLQ-C30= European Organization for Research and Treatment of Cancer Quality of Life Questionnaire-Core 30; EORTC QLQ-MY20= European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Multiple Myeloma Module 20; FACIT Item GP5=Functional Assessment of Chronic Illness Therapy Item GP5; ICANS=immune effector cell-associated neurotoxicity syndrome; IgA=immunoglobulin A; IgD=immunoglobulin D; IgE=immunoglobulin E; IgG=immunoglobulin G; IgM=immunoglobulin M; IMWG=International Myeloma Working Group; MM=multiple myeloma; MRD=minimal residual disease; NK=natural killer; ORR=overall response rate; OS=overall survival; PFS=progression-free survival; PFS2=progression-free survival after initiation of treatment for MM; PGIC=Patient Global Impression of Change; PGIS=Patient Global Impression of Severity; PK=pharmacokinetic; RNA=ribonucleic acid; SAE=serious adverse event; SLiM=greater than or equal to Sixty percent clonal plasma cells in the bone marrow, involved/uninvolved free Light chain ratio of ≥100 with the involved FLC being ≥100 mg/L, MRI with >1 focal lesion; SMM=smoldering multiple myeloma; TEAE=treatment-emergent adverse event

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Table 2: SLiM CRAB Diagnostic Criteria for a Myeloma-Defining Event

Multiple Myeloma-Defining Biomarkers	
S	Greater than or equal to **S**ixty percent clonal plasma cells in the bone marrow
Li	Involved/uninvolved free **Li**ght chain ratio of ≥100 with the involved FLC being ≥100 mg/L
M	**M**RI with >1 focal marrow lesion (at least 5 mm in size)
End Organ Damage	
C	Hyper**C**alcemia (serum calcium >1 mg/dL above ULN or >11 mg/dL)
R	**R**enal insufficiency (creatinine >2 mg/dL or creatinine clearance <40 mL/min) (Appendix 3)
A	**A**nemia (hemoglobin >2 g/dL below LLN or <10 g/dL)
B	Presence of lytic **B**one lesions (presence of one or more osteolytic lesions seen on skeletal radiography, whole-body MRI, or whole-body FDG-PET/CT)

Source: (Rajkumar, 2014) (The Myeloma Clinical Trials, 2021)
FDG-PET/CT=^{18}F-fluorodeoxyglucose positron emission tomography/computed tomography; FLC=free light chain; LLN=lower limit of normal; MRI=magnetic resonance imaging; ULN=upper limit of normal

3. HYPOTHESIS AND RATIONALE

3.1. Hypotheses

Linvoseltamab can induce durable clearance of clonal plasma cells as measured by serial assessment of CR and MRD in patients with high-risk SMM, sparing them the morbidity associated with progression to MM and toxicity related to conventional MM therapy. No formal hypothesis testing is planned in this proof-of-concept study.

3.2. Rationale

3.2.1. Rationale for Study Design

This is a phase 2 open-label study designed to assess the safety and efficacy of linvoseltamab (anti-BCMA x anti-CD3 bispecific antibody) in participants with SMM who are at a high risk of progression to multiple myeloma. As this is the first time that linvoseltamab will be evaluated in this population, enrollment will begin with a safety run-in part to assess the suitability of the proposed dosing regimen. After preliminary safety and tolerability are established, an expansion part will open to confirm the safety and evaluate the efficacy of single-agent linvoseltamab.

To limit exposure of participants to linvoseltamab if monotherapy treatment does not provide a minimum threshold of benefit, 2 interim analyses will be conducted to assess futility. The interim analyses will occur after approximately 15 and 30 participants have received 3 cycles of treatment (including participants from the safety run-in).

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3.2.2. Rationale for B Cell Maturation Antigen-Directed Therapy

High-risk SMM represents early malignancy, and safe, well-tolerated treatment strategies may prevent severe morbidity associated with progression to MM.

BCMA, a member of the tumor necrosis factor superfamily, promotes survival and proliferation of plasma cells through signal transduction of the BAFF/BLyS and a proliferation-inducing ligand. The selective expression of BCMA on mature B cells and plasma cells (neoplastic cells, in particular) has made it an attractive therapeutic target for the management of plasma cell disorders, specifically MM. Although there are not formal studies investigating BCMA expression in the bone marrow of patients with SMM, it is known that soluble BCMA levels are elevated in patients with SMM and higher in SMM patients who progress to MM (Visram, 2021). This background knowledge provides motivation to evaluate BCMA-targeted therapy in high-risk SMM patients.

There are a variety of approaches in use to target BCMA: antibody drug conjugates, BCMAxCD3 bispecific monoclonal antibodies or other T-cell engagers, and CAR T cells. BCMAxCD3 bispecific monoclonal antibodies like linvoseltamab are attractive therapeutic agents given their off-the-shelf availability, encouraging efficacy, and manageable safety profile. Currently, teclistamab is the only anti-BCMA bispecific antibody approved in the EU.

3.2.3. Rationale for Linvoseltamab Treatment in Patients with Smoldering Multiple Myeloma

Smoldering MM is an intermediate asymptomatic condition that lies between MGUS and MM along the spectrum of plasma cell dyscrasias. It follows an increasingly predictable progression pattern according to various clinical and molecular biomarkers (Landgren, 2021) (Rajkumar, 2022). Interception before the onset of frank malignancy represents an attractive strategy to actively intervene when patients are healthier without evidence of end-organ damage and potentially more responsive to therapy, which may lead to better overall outcomes. Early therapy with lenalidomide has already been shown to be highly effective at preventing progression from high-risk SMM to MM (Lonial, 2020) (Mateos, 2013).

Linvoseltamab (REGN5458) is an anti-BCMA x anti-CD3 bispecific antibody that binds to BCMA on plasma cells and to CD3 on T cells. In vitro experiments with linvoseltamab have demonstrated activation of Jurkat T lymphocytes exclusively in the presence of human BCMA-expressing cells and target-mediated cytotoxicity in a range of MM cell lines and primary MM cells with variable BCMA cell surface expression. Additional details regarding preclinical data are provided in the linvoseltamab IB. The phase 1 results from RRMM indicate that linvoseltamab is tolerable with a manageable safety profile. Cytokine release syndrome is the most common TEAE. Most CRS events are grade 1 and occur shortly after administration of the first or second step-up dose, typically resolving within 24 hours. The overall incidence of grade ≥3 treatment-related AEs was 42.7%, with the most common being anemia and neutropenia (Zonder, 2021). Encouraging efficacy has been reported in participants that received a median of 5 prior lines of therapy, with 89% being triple-class refractory or worse. Most responses occurred within the first month of treatment and deepened over time. Among responders across all dose groups, 86% (n=32/37) achieved a VGPR or better, 43% (n=16/37) achieved a complete response (CR), with 40% of evaluable CR patients (n=4/10) being minimal residual disease (MRD) negative. The probability of an ongoing response at 8 months was 90.2%. Detailed information about the reported depth, duration, and durability of responses is provided in the linvoseltamab IB.

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Based on the above background and ongoing development of aggressive multiagent treatment regimens in high-risk SMM (Kazandjian, 2021) (Mateos, 2021), the available data suggest that the use of single-agent linvoseltamab may reasonably yield similar or better efficacy outcomes and less toxicity compared to other emerging treatments. Therefore, the safety and efficacy of linvoseltamab in patients with SMM at high risk of progression to MM will be investigated. Since this is the first time that linvoseltamab will be evaluated in this population, the study will begin with a safety run-in cohort of 6 participants, followed by an expansion cohort of up to 34 participants.

3.2.4. Rationale for Linvoseltamab Dosing Regimen

Selection of the linvoseltamab dosing regimen in this study is informed by the biological similarity between high-risk SMM and MM as well as the cumulative safety and efficacy data from the ongoing FIH study (R5458-ONC-1826) in which linvoseltamab is being investigated as monotherapy in participants with RRMM. Detailed clinical data are provided in the linvoseltamab IB.

3.2.4.1. Rationale for Step-Up Schedule and Inpatient Monitoring

Linvoseltamab, like other T cell-redirecting therapies, can induce CRS during the first few weeks of treatment, and priming the T-cell response with stepwise dose escalation (step-up dosing) has been shown to reduce the risk of severe CRS. Across the higher full-dose levels (96 to 800 mg) studied, a step-up schedule consisting of an initial 5 mg dose followed a week later by an intermediate 25 mg dose adequately mitigated the risk of CRS in conjunction with premedications and inpatient monitoring for 24 hours after the end of infusion. As of 30 Sept 2022, clinical data from R5458-ONC-1826 indicate an overall favorable CRS profile characterized by a low rate of grade 2 or higher CRS (grade 2: 22/218 [10.1%]; grade 3: 3/218 [1.4%]) and relatively rapid onset and resolution of CRS events. The median time to onset of CRS was 11.0 hours from the start of infusion, and the median duration was 15.6 hours. No cases of grade ≥2 CRS began after approximately 24 hours from the end of infusion, and only 2 of 218 participants (0.9%) experienced grade ≥2 CRS after receiving the first full dose (both grade 2). These results support the adequacy of precautionary inpatient monitoring for 24 hours after each step-up dose. Additionally, given there is less urgency to achieve an immediate therapeutic response in SMM, a more gradual step-up schedule may further mitigate the risk of CRS and obviate the need for inpatient monitoring. The current study will therefore implement a step-up schedule that splits the 5 mg dose (initial step-up dose of linvoseltamab in RRMM) into 1 mg at week 1 and 4 mg at week 2 followed by 25 mg at week 3 before reaching the full dose of linvoseltamab.

3.2.4.2. Rationale for Full Dose Selection

Participants in the FIH study have tolerated up to a weekly full dose of 800 mg without an identified maximum tolerated dose. The incidence rate of TEAEs, SAEs, and TEAEs grade ≥3 has been similar across all dose levels and there is no apparent sign of dose-dependency, albeit longer follow-up is needed. Although the most common non-hematologic TEAE associated with linvoseltamab was CRS (44.2%), no correlation between the severity of CRS and the full dose level (up to 800 mg) was observed. The most frequently reported grade ≥3 hematologic TEAEs were neutropenia, anemia, lymphopenia, and thrombocytopenia. Additional clinical safety data are provided in the linvoseltamab IB.

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Linvoseltamab has never been studied as a treatment for high-risk SMM patients. However, efficacy data from the phase 1 portion of study R5458-ONC-1826 indicate a trend towards higher response rates at higher doses, with ORRs of 29% (7/24), 48% (12/25), and 75% (18/24) at the dose levels of 3 to 12 mg, 24 to 96 mg, and 200 to 800 mg, respectively. The ORR across the dose range of 200 to 800 mg appeared to remain stable with estimates of 83.3% (10/12), 62.5% (5/8) and 75% (3/4) at 200, 400, and 800 mg, respectively. Detailed information about the depth, duration, and durability of responses is provided in the linvoseltamab IB.

The totality of data from R5458-ONC-1826 indicates that a full dose of 200 mg linvoseltamab is likely to provide participants with favorable clinical benefit while maintaining a 4-fold margin in dose level below the highest dose (800 mg QW) tested thus far. Given the biological similarity between MM and high-risk SMM, the dose of 200 mg linvoseltamab is therefore proposed for investigation as monotherapy in this proof-of-concept study for high-risk SMM patients.

3.2.4.3. Rationale for Dosing Schedule

In the phase 2 portion of the FIH study, after completion of the step-up period, treatment with full-dose linvoseltamab adhered to the following schedule of administration: QW for 12 weeks, then Q2W for at least 8 weeks, and then Q4W if the participant reaches a VGPR or better until progression. Using this regimen, most RRMM patients achieved a meaningful clinical response within the first 4 to 8 weeks of treatment (2 to 6 full doses) that deepened over time. Implementation of a less intensive dosing regimen (QW for 4 weeks, then Q2W for 16 weeks, then Q4W for up to 2 years) is proposed in this study to better optimize the dosing schedule for the indication of SMM. A linvoseltamab dose of 200 mg given every 4 weeks is predicted to maintain a median C_{trough} concentration above where complete responses have been observed and maintained in patients with RRMM. Furthermore, the generally lower tumor burden and more intact immune system found in patients with high-risk SMM may enable superior efficacy at a lower dose intensity while preventing T-cell exhaustion (Goebeler, 2020) (Philipp, 2022).

Fixed-duration treatment with linvoseltamab is desirable to prevent overtreatment and exposure to excess risk from prolonged immunosuppression. Data from the two phase 3 clinical trials that demonstrated the benefits of early therapy with lenalidomide in high-risk SMM patients support this strategy (Lonial, 2020) (Mateos, 2013). Both studies were designed to treat until progression, but the Spanish study was amended to limit the duration of treatment to 2 years due to the risk of second primary malignancies identified in MM studies using lenalidomide. Given the comparable outcomes and desire to limit long-term toxicity, current guidelines recommend 2 years of therapy for newly diagnosed high-risk SMM patients (NCCN, 2022) (Rajkumar, 2022). Of note, most high-risk SMM patients treated with lenalidomide achieve no better than a partial response, yet the limited duration of therapy still provides a significant PFS advantage compared to observation. Much deeper responses are expected after 2 years of treatment with linvoseltamab, suggesting that long-lasting clinical benefit may be observed. Indeed, RRMM patients have demonstrated sustained responses for more than 1 year after discontinuation of linvoseltamab. Therefore, the proposed dosing schedule is intended to improve convenience and potentially reduce long-term risks like infection without sacrificing anticipated efficacy, resulting in a more favorable risk-benefit proposition for a healthier patient population compared to RRMM.

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3.2.5. Rationale for Exploratory Objectives

The exploratory analysis plan in high-risk SMM will include evaluation of the mechanism of action of linvoseltamab to induce T-cell activation and plasma cell depletion. Given that this patient population is different than RRMM, the dose dependent engagement of T cells in this population may inform identification of an appropriate dose regimen. In peripheral blood, linvoseltamab may induce changes in T cells and B cells that are important to characterize to inform the dose selection and to inform tolerability in this patient population.

In this study, detection and phenotyping of circulating plasma cells will be evaluated at baseline and post-treatment to inform plasma cell depletion anticipated by linvoseltamab treatment. Monitoring circulating plasma cells and detection of BCMA expression as well as cytogenetics has been reported for SMM patients (Ndacayisaba, 2022). As characterization of BCMA expression in bone marrow of high-risk SMM patients has not been reported, analysis of bone marrow core biopsies for BCMA expression will be performed.

3.2.6. Rationale for Patient-Reported Outcomes

SMM is an asymptomatic condition that has historically been monitored without treatment. Maintenance of QoL is an important consideration when assessing the risk-benefit of introducing a potentially toxic treatment. The EORTC-QLQ-C30 and EORTC-QLQ-MY20 are established health questionnaires that have been used extensively within the setting of MM. Notably, interventional trials with daratumumab (NCT03301220) and isatuximab (NCT04270409) are already evaluating these PROs within the high-risk SMM population. The FACIT Item GP5 is recommended by the FDA for evaluating patient-reported global impact of treatment side effects in cancer trials.

All PRO endpoints are exploratory.

3.3. Risk-Benefit

3.3.1. Risk Assessment of Linvoseltamab

Important risks of clinical significance, rationale for risks, and associated mitigation strategies are presented in Table 3. Important risks include identified and potential risks (see the linvoseltamab IB for additional details).

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Table 3: Risks Assessment of Linvoseltamab (REGN5458)

Important Risks of Clinical Significance	Rationale for Risk	Mitigation Strategy
CRS/IRR/systemic hypersensitivity reactions	There is significant overlap in the clinical presentation of CRS, IRR, and systemic hypersensitivity. The symptomatology associated with these entities is consistent with an underlying mechanism of exaggerated immune response and release of inflammatory cytokines. The mechanism of action of T cell-engaging immunotherapy such as linvoseltamab is non-physiologic T-cell activation with the occurrence of cytokine release. Clinical effects of these can range from mild symptoms, such as fever, to a severe inflammatory syndrome, which may be life-threatening or lead to death.	Participants at high risk of hypersensitivity reactions are excluded (Exclusion Criterion #6). Medications and equipment to manage medical emergencies must be available for immediate use (Section 8.3.1 and Section 8.3.2). Stepwise increase in dose (step-up period) prior to reaching full dose is used (Section 6.1.1, Section 6.1.2, and Section 6.1.3.1). Infusion of linvoseltamab in a monitored setting and subsequent inpatient monitoring following administration of initial and intermediate doses for early detection and management of CRS/IRR and hypersensitivity is required (Section 6.1.1 and Section 8.3). Premedication with dexamethasone, antihistamines, and paracetamol (acetaminophen) is required for all participants from step-up doses through first full dose (Section 8.1.2). Guidance to investigators on drug interruption and/or termination, grading, and management of symptoms of CRS/IRR and hypersensitivity reactions is provided in Section 8.3.1 and Section 8.3.2.
ICANS	CNS symptoms such as confusion and aphasia have been observed with CAR T cell therapy and other bispecific antibodies and were previously considered part of CRS. These symptoms are now considered to be a separate syndrome called ICANS. Cytokines may be implicated in the pathophysiology (Lee, 2019).	Participants with CNS pathology are excluded (Exclusion Criterion #13). Guidance for early identification of neurologic events and monitoring is provided in Section 8.3.3. Guidance on management of neurological toxicity including study drug administration is provided in Table 11.

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Important Risks of Clinical Significance	Rationale for Risk	Mitigation Strategy
Serious Infections	Based on the mechanism of action, treatment with linvoseltamab may result in pronounced B-cell depletion and hypogammaglobulinemia. The expected primary pharmacology of B-cell depletion may result in increased risk of infections, including opportunistic infections and viral infections (including HBV and CMV).	Participants with known active infections are excluded (Exclusion Criteria #4 and #5). Immunoglobulin levels will be checked at baseline and levels will be monitored periodically throughout the trial (Section 9.1). Total lymphocyte counts and T lymphocyte subsets will be checked at baseline and monitored periodically throughout the trial (Section 9.1). Consideration of prophylaxis for viral infections (herpesviruses) and fungal infection, and intravenous immunoglobulin for participants with severe hypogammaglobulinemia, is recommended (Section 8.3.4).
Tumor Lysis Syndrome	Tumor lysis syndrome is an oncologic emergency triggered by the rapid release of intracellular material from lysing malignant cells. Tumor lysis syndrome has been reported in virtually every cancer type. The highest risk of TLS is seen in large-volume, highly metabolic malignancies such as B cell acute lymphoblastic leukemia and Burkitt's lymphoma, whereas solid tumors and slow-growing hematologic malignancies (such as MM) carry lower risks (Wilson, 2014).	Participants with hypersensitivity to both allopurinol and rasburicase are excluded (Exclusion Criterion #7). Section 8.3.5 provides guidance on prophylaxis, monitoring, and management of TLS
Embryo-Fetal Toxicity	Reproductive and developmental toxicology studies have not been conducted; therefore, the effects of linvoseltamab on reproduction and embryo-fetal development are unknown.	Pregnancy testing is performed for women of childbearing potential (Inclusion Criterion #8.b). Pregnant or breastfeeding women are excluded (Inclusion Criterion #8.b). Contraception is required to prevent pregnancy in study participants or their sexual partners (Inclusion Criterion #8).

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Important Risks of Clinical Significance	Rationale for Risk	Mitigation Strategy
Clinical consequences of immunogenicity: ADAs	All protein therapeutics, including linvoseltamab, have the potential to induce an immunogenic response (formation of antibodies against the study drug) following administration.	Anti-linvoseltamab antibody status will be assessed during the study (Table 14). Evaluation of clinical consequences of immunogenicity will be completed (Section 2).

ADAs=anti-drug antibodies; CAR=chimeric antigen receptor; CNS=central nervous system; CRS=cytokine release syndrome; HBV=hepatitis B virus; ICANS=immune effector cell-associated neurotoxicity syndrome; IRR=infusion-related reaction; MM=multiple myeloma; TLS=tumor lysis syndrome

More information about the known and expected benefits and risks of linvoseltamab may be found in the most recent version of the linvoseltamab IB.

3.3.2. COVID-19 Considerations

Recognizing that an event of unforeseen circumstances (eg, COVID-19 or other national/international crisis) may have an impact on the conduct of clinical trials, the sponsor does not intend to screen any participants in this study until the impact of the emergency is deemed manageable and no longer interfering with the conduct of studies at individual sites, and patients can safely participate in this study. For information regarding the continuity of clinical study conduct in unforeseen circumstances, see Section 9.1.

For information regarding the permitted timing of COVID-19 vaccinations, see Section 7.2.2 criterion #14 and Section 8.7.

3.3.3. Benefit Assessment of Linvoseltamab

Linvoseltamab has demonstrated promising anti-tumor activity in heavily pretreated patients with RRMM. Responses, including ≥VGPR, were observed at all dose levels studied in the FIH study with an apparent plateau in efficacy to support further evaluation of the recommended 200 mg full dose (see Section 3.2.4).

Additional data are needed to better understand the clinical benefit of linvoseltamab in participants with high-risk SMM. See the most recent version of the linvoseltamab IB for additional details.

3.3.4. Overall Benefit-Risk Conclusion

The available nonclinical and preliminary clinical data from the ongoing FIH study indicate linvoseltamab has a manageable safety profile and provides encouraging anti-tumor benefit to participants with RRMM. The linvoseltamab IB provides a risk-benefit statement with respect to the overall development program.

Considering the multiple safety measures taken to minimize important risks of linvoseltamab, and the anticipated benefits that may be afforded to participants with high-risk SMM participating in this study, clinical development of linvoseltamab in participants with high-risk SMM is justified.

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4. ENDPOINTS

See Table 1 for endpoints.

5. STUDY VARIABLES

5.1. Demographic and Baseline Characteristics

Baseline characteristics will include standard demography (eg, age, race, etc), physical examination findings (including height and weight), and medical history and oncologic history (including prior cancer-related therapies, radiotherapies, and surgeries) for each participant.

5.2. Efficacy Variables

Efficacy variables include, but are not limited to, the following:

- Best overall response (PR or better) as measured by IMWG criteria (Appendix 1)
- DOR by IMWG criteria (Appendix 1) among participants who achieve a response of PR or better
- PFS as measured by IMWG criteria (Appendix 1)
- MRD negative status (at 10^{-5})
- OS

5.3. Safety Variables

Safety variables include, but are not limited to, the following:

- Vital signs, as described in Section 9.2.3.2
- Physical examination, as described in Section 9.2.3.3
- Neurologic examination results, as described in Section 9.2.3.4
- Clinical laboratory results, as described in Section 9.2.3.6
- AESIs, as described in Section 10.1.3 and Section 10.2.3
- AEs, as described in Section 10.2.1
- SAEs, as described in Section 10.2.2

5.4. Pharmacokinetic Variables

The PK variable is the concentration of total linvoseltamab in serum over time. These sampling timepoints are specified in Table 15.

5.5. Immunogenicity Variables

The immunogenicity variables are ADA status, titer, and time-point/visit. Samples in this study will be collected at the clinic visits specified in Table 15.

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5.6. Pharmacodynamic and Other Biomarker Variables

Biomarker variables include the percentages of T, B, and NK cells, monocytes, and CTCs over time in peripheral blood. Target engagement by linvoseltamab includes evaluation of the percentages of activated T and NK cells in peripheral blood over time assessed by markers such as CD69 and CD25, as well as induction of serum cytokines such as IL-2, IL-6, IL-10, IFN-gamma, and TNF-alpha. Treatment may impact normal plasma cell survival, therefore the concentrations of serum polyclonal immunoglobulin IgG, IgA, IgE, and IgM (excluding the clonal subtype) will be evaluated over time. Other biomarkers related to plasma cell disorders or the mechanism of action of linvoseltamab may be evaluated, including sBCMA. Exploratory biomarkers are described in Section 9.2.7.

5.7. Patient-Reported Outcome Variables

PRO variables include patient-reported symptoms, functioning, and general health status/QoL as measured by validated instruments EORTC QLQ-C30 (Aaronson, 1993) and EORTC QLQ-MY20 (Cocks, 2007) (Stead, 1999). Two global anchors, the PGIC and PGIS, will be used to aid in the interpretation of EORTC QLQ-C30 and EORTC QLQ-MY20 (Gary, 2020) (Guy, 1976). Global impact of side effects will be measured by FACIT Item GP5.

5.7.1. Variables from European Organization for Research and Treatment of Cancer Quality of Life Questionnaire - Core 30

The EORTC QLQ-C30 scale scores will be calculated and missing data will be handled per developer instructions using the EORTC QLQ-C30 Scoring Manual (Fayers, 2001). The instrument yields the following scales:

- GHS/QoL with 2 items
- Functional scales
  - PF with 5 items
  - RF with 2 items
  - EF with 4 items
  - CF with 2 items
  - SF with 2 items
- Symptom scales/items
  - FA with 3 items
  - NV with 2 items
  - PA with 2 items
  - DY, SL, AP, CO, DI with 1 item each
- FI with 1 item

Change from baseline in EORTC QLQ-C30 scale scores (transformed 0 to 100 scores) will be categorized as improvement/stable/deterioration using a 10-point threshold value based on

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available literature (Osoba, 1998) (Kvam, 2010) and global anchors of PGIS and PGIC (Appendix 2).

5.7.2. Variables from European Organization for Research and Treatment of Cancer Quality of Life Questionnaire-Multiple Myeloma Module 20

The EORTC QLQ-MY20 instrument yields the following scales:

- DS with 6 items
- SE with 10 items
- FP with 3 items
- BI with 1 item

Change from baseline in EORTC QLQ-MY20 scale scores (transformed 0 to 100 scores) will be categorized as improvement/stable/deterioration using clinically meaningful thresholds identified based on literature (Sully, 2019) and global anchors of PGIS and PGIC.

5.7.3. Variables from FACIT Item GP5

FACIT Item GP5 is a single-item ("I am bothered by side effects of treatment") summary measure of patient-reported overall impact of treatment toxicity measured and scored on a 5-point Likert-type scale (0=Not at all, 1=A little bit, 2=Somewhat, 3=Quite a bit, 4=Very much).

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6. STUDY DESIGN

6.1. Study Description and Duration

This is an open-label, phase 2 study designed to evaluate the safety and efficacy of linvoseltamab in participants with SMM at high risk of progression to MM. As this is the first time that linvoseltamab will be evaluated in this population, enrollment will begin with a safety run-in part to assess the suitability of the recommended dosing regimen. After early safety and tolerability are established, an expansion part will open to confirm the safety and evaluate the preliminary clinical activity of single-agent linvoseltamab.

6.1.1. Description of Study Cohorts

Safety Run-In (Part 1): At least 6 participants will receive linvoseltamab monotherapy starting with a step-up schedule that consists of weekly dose increments of 1 mg (week 1), 4 mg (week 2), and 25 mg (week 3) before reaching a full dose of 200 mg (Figure 1). Each participant must be monitored in an inpatient setting for at least 24 hours after each step-up dose infusion as a precautionary safety measure. The step-up period is expected to last 3 weeks unless a tolerability issue requires repeat administration of step-up doses. All subsequent treatment will be given over 28-day cycles. Full-dose linvoseltamab will be administered weekly (QW) for the first cycle and then every 2 weeks (Q2W) for the next 4 cycles. At the start of cycle 6, the dosing frequency will decrease to every 4 weeks (Q4W) and continue until completion of 24 cycles or fulfillment of a protocol-defined criterion for study drug discontinuation (Figure 2).

Participants enrolled as part of the safety run-in must start treatment with linvoseltamab in a staggered fashion (no less than 48 hours apart) to allow sufficient observation for acute toxicities and limit risk to other participants if such events are detected. The safety run-in observation period will last 35 days for each participant, beginning on the day when the first step-up dose is administered. Tolerability will be assessed by the incidence of AESI and grade ≥3 AEs.

Based on the data collected and reviewed, the SSET may recommend modification to the dosing regimen (eg, adjustment of step-up dosing). Depending on this analysis, additional cohorts with different regimens could be introduced by protocol amendment.

Expansion (Part 2): An additional 34 participants will be enrolled to ensure at least 40 participants receive linvoseltamab monotherapy (Figure 1) according to the same dosing regimen established in the safety run-in part (Figure 2). The expansion cohort will open after careful review of all the safety and efficacy data gathered from the 6 participants treated in the safety run-in cohort.

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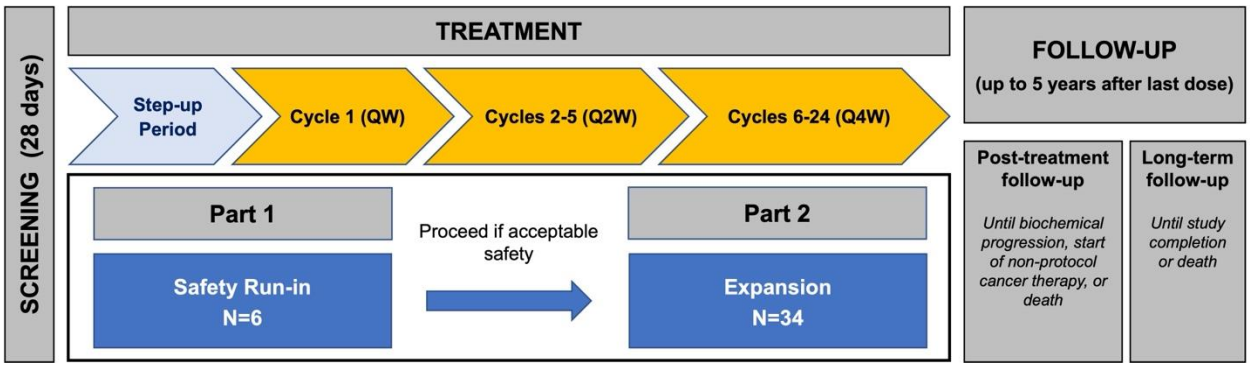
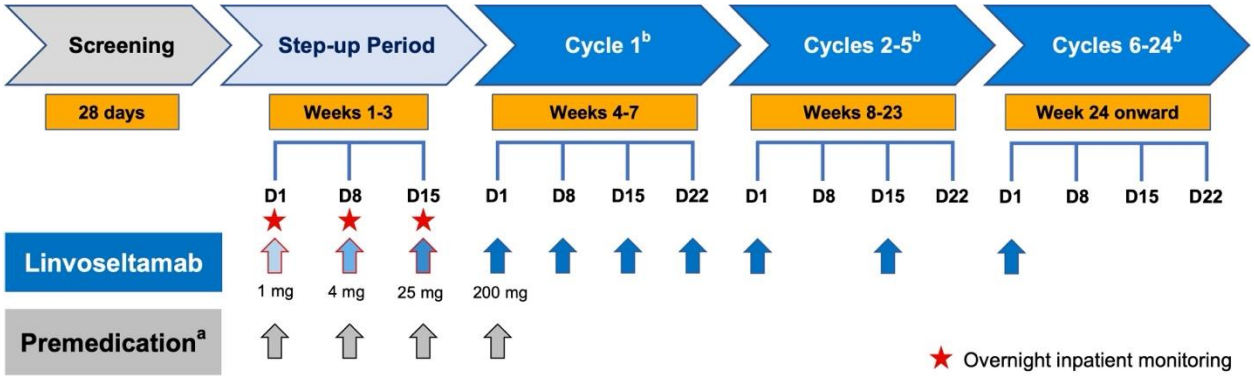
Figure 1: Study Flow Diagram



QW=weekly; Q2W=every 2 weeks; Q4W=every 4 weeks

Figure 2: Study Dosing Schema



[a] See Section 8.1.2 for premedication (dexamethasone, acetaminophen, antihistamine) requirements.
[b] Weekly administration of linvoseltamab will occur during the first cycle of treatment and then transition to Q2W for the next 4 cycles. At the start of cycle 6, the dosing frequency will be decreased to Q4W. Treatment may continue for up to 2 years.

6.1.2. Study Duration

For each participant, the study will span the following 3 periods:

- **Screening Period:** Screening procedures may occur over a period of up to 28 days for each study participant, ending on the day preceding the planned first dose of linvoseltamab.
- **Treatment Period:** The treatment period starts on the day when the first dose of linvoseltamab is administered and continues until completion of the end of treatment visit or, if the end of treatment visit does not occur, 30 days after the last dose of linvoseltamab is administered. Treatment may continue for up to 24 cycles or until fulfillment of a protocol-defined criterion for study drug discontinuation.
- **Follow-Up Period (up to 5 years after last dose):**
  - **Post-treatment follow-up:** Post-treatment follow-up starts after the end of treatment visit and involves ongoing collection of safety and efficacy data. Safety monitoring will continue for up to 90 days after the last dose of linvoseltamab.

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Participants with an ongoing biochemical response will continue to undergo routine efficacy assessments until confirmed biochemical progression, start of non-protocol cancer therapy, or death.

– **Long-term follow-up:** Long-term follow-up begins after post-treatment follow-up to collect survival status as well as other relevant long-term outcomes (eg, progression to MM, subsequent non-protocol cancer therapy, and stem cell collection) and will continue for up to 5 years after last dose of study drug, unless death or study termination by the sponsor occurs earlier. Participants may be followed remotely for survival and subsequent treatment history.

6.1.3. Study Safety and Stopping Rules

6.1.3.1. Safety Run-In Observation Period

The safety run-in will evaluate the suitability of a 3-dose step-up regimen (1 mg, 4 mg, and 25 mg) leading up to the 200 mg full dose of linvoseltamab. All participants included in the safety run-in will be observed for 35 days from the start date of the first linvoseltamab infusion. A participant will be considered evaluable during the safety run-in if they 1) receive at least 1 full dose (200 mg) of linvoseltamab and complete the 35-day observation period, or 2) experience a treatment-related AESI or grade ≥3 AE prior to completing the 35-day observation period (see excess toxicity criteria below). If the full dose is not reached due to reasons other than toxicity (eg, disease progression, missed appointments, non-compliance, participant withdrawal), the participant may be replaced with a new participant at the discretion of the sponsor.

In addition to the trigger event thresholds (Section 6.1.3.3) that will be continuously monitored throughout the entire duration of study treatment, the occurrence of AESI and grade ≥3 AEs will be closely monitored during the 35-day safety run-in observation period. Toxicity potentially related to study treatment that exceeds the limits described below will prompt an enrollment pause pending full review of available clinical data and reassessment of the linvoseltamab dosing regimen (step-up and full dose) by the SSET. Excess toxicity is defined as any of the following:

- Combined rate of grade 2 CRS and ICANS exceeding 20%
- Grade ≥3 CRS or ICANS experienced by ≥1 participant
- Grade ≥3 treatment-related AE experienced by ≥2 participants, except for:
  - Grade 3 laboratory abnormalities without clinical consequence
  - Grade 3 laboratory abnormalities transiently associated with grade ≤2 CRS

On condition of satisfying the above criteria and SSET agreement, the dosing regimen will be deemed acceptable for further study in the expansion part. The SSET will evaluate all available data collected during the 35-day observation period including safety data from inevaluable participants. Toxicities observed beyond the safety run-in observation period will also be considered. Alternative dosing regimens may be explored depending on safety and tolerability. If a change is deemed necessary, revisions to the dosing regimen and rationale will be described in a protocol amendment.

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6.1.3.2. Study Safety Evaluation

The SSET is responsible for overseeing the safety run-in period and start of expansion. Formal meetings will be conducted regularly including at the end of the safety run-in and when required on an ad hoc basis. Members of the SSET will include representatives from the sponsor's clinical study team (the medical monitor, clinical study team lead, study biostatistician, and global patient safety lead) and the clinical trial principal investigators. Information reviewed by the SSET will be comprehensive and include participant demographics, medical/oncology history, frequency and severity of AEs, clinical and specialty laboratory data, and available clinical PK or pharmacodynamic data. A final decision on whether to proceed with the expansion part of the study will ensue after thorough discussion and be contingent upon the consensus judgment of the SSET.

The sponsor will continue to monitor cumulative safety data throughout the entire treatment period for all participants (safety run-in and expansion).

6.1.3.3. Conditions for Enrollment Pause and Safety Review

Participants will be monitored for trigger events over the entire treatment period, starting with the first infusion of linvoseltamab. These trigger events will include all cases of grade ≥3 CRS, grade ≥3 ICANS, grade ≥3 treatment-related infection, and grade ≥4 treatment-related AEs (except grade 4 lymphopenia). A stopping bound is set based on the lower bound of the exact one-sided 80% CI calculated using the Clopper-Pearson method. Thresholds have been defined for the cumulative rate of all trigger events, and then for specific AEs. For all trigger events, if the lower bound exceeds 20%, then enrollment will be paused. Threshold examples are shown in Table 4.

Table 4: Stopping Boundaries to Pause Enrollment for Cumulative Rate of All Trigger Events

Number of participants treated	Number of participants with trigger events leading to pause
2-4	≥ 2
5-7	≥ 3
8-11	≥ 4
12-15	≥ 5
16-20	≥ 6
21-24	≥ 7
25-28	≥ 8
29-32	≥ 9
33-37	≥ 10
38-40	≥ 11

For grade ≥3 treatment-related infection and grade ≥4 non-hematologic AEs, if the lower bound exceeds 10%, then enrollment will be paused (Table 5).

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Table 5: Stopping Boundaries to Pause Enrollment for Cumulative Rate of Treatment-Related Grade ≥3 Infection and Grade ≥4 Non-Hematologic AEs

Number of participants treated	Number of participants with trigger events leading to pause
3-8	≥ 2
9-15	≥ 3
16-23	≥ 4
24-31	≥ 5
32-39	≥ 6
40	≥ 7

Enrollment will also be immediately paused for any of the following:

- Grade 4 CRS or ICANS
- Treatment-related death

During a study pause, accrual will be suspended and relevant regulatory authorities will be notified. The sponsor will review all available data for participants treated with linvoseltamab and re-evaluate the dosing regimen and dose selection. Remedial actions such as dose reduction will be considered based on the overall risk-benefit assessment.

6.1.3.4. Individual Patient Stopping Rules

Reasons for permanent discontinuation of study drug are listed in Section 8.2.3.1.

6.1.4. End of Study Definition

An individual participant will reach the end of study whenever one of the following events occurs: completion of 5 years of follow-up after the last dose of study drug, death, loss to follow-up, or withdrawal of consent.

The study/trial will end on the date of the last participant's last visit for any of the reasons listed above or study termination by the sponsor, whichever occurs earlier.

6.2. Planned Interim Analyses

Interim analyses for futility will be performed after 15 and 30 participants have started treatment with linvoseltamab and completed at least 15 weeks of follow-up with at least one post-baseline response assessment.

The futility criteria for the first interim analysis are observing 5 or fewer responses or no CR.

The futility criteria for the second interim analysis are observing 13 or fewer responses or 2 or fewer CRs.

Enrollment will not be paused during interim analyses.

A description of the statistical methods to be employed is in Section 11.5, and blinding implications are discussed in Section 8.5.

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6.3. Study Committees

A formal committee is not required. The sponsor may elect to obtain independent reviews of the efficacy data. The process for such reviews would be set forth in a separate charter.

The decision to stop the study prematurely due to safety prior to the end of study will be made by the SSET (Section 6.1.3.2).

7. SELECTION, WITHDRAWAL, AND REPLACEMENT OF PARTICIPANTS

7.1. Number of Participants Planned

Approximately 40 participants are planned in this study: at least 6 in the safety run-in (Part 1) and 34 in the expansion (Part 2).

7.2. Study Population

The study population is adult patients with SMM at high risk of progression to MM.

7.2.1. Inclusion Criteria

A participant must meet the following criteria to be eligible for inclusion in the study:

1. Age ≥18 years at the time of signing the informed consent.
2. SMM diagnosis within 5 years of study enrollment per IMWG criteria defined as:
   a. Serum M-protein ≥30 g/L **or** urinary M-protein ≥500 mg per 24 hours **or** BMPCs 10% to <60%

      AND

   b. Absence of myeloma-defining events or other related conditions (SLiM CRAB)
3. High-risk SMM according to the following markers at screening:
   a. At least 2 of the following abnormalities
      – Serum M-protein >20 g/L or 2 g/dL
      – Serum involved/uninvolved FLC ratio >20
      – BMPCs >20%

      AND/OR

   b. Both of the below findings (PETHEMA criteria):
      – ≥95% aberrant/clonal BMPCs
      – Immunoparesis defined as reduction of ≥1 uninvolved Ig isotype of at least 25% below the lower normal limit (only IgG, IgA, and IgM will be considered)
4. ECOG performance status ≤1
5. Adequate hematologic function:

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a. Absolute neutrophil count ≥1500/μL ($1.5x10^9$/L). No G-CSF is permitted to reach this value during the screening period.
b. Platelets ≥75,000/μL ($75x10^9$/L) (not permissible to transfuse a participant within 2 weeks prior to screening to reach this level).
c. Clinically stable anemia that is clearly attributable to a condition other than MM (eg, thalassemia, iron deficiency) and that has not required transfusions in the prior year may be considered for inclusion after discussion with the medical monitor.

6. Adequate hepatic function:
   a. Total bilirubin ≤2 mg/dL (except known Gilbert syndrome, in which case indirect bilirubin should be ≤5 mg/dL)
   b. Transaminases (ALT/AST) ≤3 x ULN
7. Estimated glomerular filtration rate ≥30 mL/min/1.73 m^2 by the MDRD equation (Appendix 3).
8. Contraception
   a. Men: A male participant must agree to use condoms or practice sexual abstinence[†, ‡] unless vasectomized prior to the first dose/start of treatment, during the study, and for at least 6 months after the last dose. Sperm donation is prohibited during the same period. Vasectomy success must be confirmed by semen analysis.

      [†] Sexual abstinence is considered a highly effective method of contraception only if defined as refraining from heterosexual intercourse during the entire period of risk associated with the study regimen. The reliability of sexual abstinence needs to be evaluated in relation to the duration of the clinical trial and the preferred and usual lifestyle of the participant.

      [‡] Periodic abstinence (calendar, symptothermal, post ovulation methods), withdrawal (coitus interruptus), spermicides only, and LAM are not acceptable methods of contraception. Female condom and male condom should not be used together.

   b. Women: A female participant must not be pregnant or breastfeeding and must have a negative pregnancy test at screening (serum β-hCG).

      WOCBP must agree to undergo pregnancy testing and practice highly effective contraception* prior to the first dose/start of treatment, during the study, and for at least 6 months after the last dose. Egg donation (ova, oocytes) for the purposes of assisted reproduction is prohibited during the same period.

      – WOCBP are defined as women who are fertile following menarche until becoming postmenopausal, unless permanently sterile. Permanent sterilization methods include hysterectomy, bilateral salpingectomy, and bilateral oophorectomy. A postmenopausal state is defined as no menses for 12 months without an alternative medical cause. An FSH level in the postmenopausal range may be used to confirm a postmenopausal state in women not using hormonal contraception or hormonal replacement therapy. However, in the absence of 12 months of amenorrhea, a single FSH measurement is insufficient to determine the occurrence of a postmenopausal state. The above definitions are according to CTFG guidance.

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*Highly effective contraceptive measures include:

- Stable use of combined (estrogen- and progestogen-containing) hormonal contraception (oral, intravaginal, transdermal) or progestogen-only hormonal contraception (oral, injectable, implantable) associated with inhibition of ovulation initiated 2 or more menstrual cycles prior to screening;
- IUD or IUS;
- Bilateral tubal ligation (also called bilateral tubal occlusion);
- Vasectomized partner (provided that the male vasectomized partner is the sole sexual partner of the WOCBP study participant, and the vasectomized partner has obtained medical assessment of surgical success); and/or
- Sexual abstinence [†, ‡]

Pregnancy testing and contraception are not required for women who are postmenopausal or permanently sterile.

9. Capable of giving voluntary written informed consent as described in the protocol.

7.2.2. Exclusion Criteria

A participant who meets any of the following criteria will be excluded from the study:

1. Evidence of any of the following myeloma defining events (SLiM CRAB) attributable to the underlying plasma cell dyscrasia:
   a. Hypercalcemia: corrected serum calcium >1 mg/dL above the ULN or >11 mg/dL
   b. Renal insufficiency: GFR <40 mL/min by MDRD equation or serum creatinine >2 mg/dL (Appendix 3)
   c. Anemia: hemoglobin 2 g/dL below LLN or <10 g/dL

      Transfusion support or concurrent treatment with erythropoietin stimulating agents is not permitted.
   d. ≥1 osteolytic bone lesion detected by low-dose CT or FDG-PET/CT (x-ray is not acceptable)
   e. BMPCs ≥60%
   f. Serum involved/uninvolved FLC ratio ≥100
   g. Whole body magnetic resonance imaging (MRI) with more than 1 focal lesion (≥5 mm in size)
2. Diagnosis of systemic light chain amyloidosis, Waldenström macroglobulinemia (lymphoplasmacytic lymphoma), soft tissue plasmacytoma, or symptomatic multiple myeloma.
3. Clinically significant cardiac or vascular disease within 3 months of study enrollment including but not limited to myocardial infarction, unstable angina, coronary revascularization (eg, coronary artery bypass graft, percutaneous coronary intervention), peripheral artery revascularization, left ventricular ejection fraction <40% by echocardiogram or MUGA scan, NYHA class III or IV heart failure, transient ischemic

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attack, stroke, thromboembolic event (eg, pulmonary embolism), or unstable cardiac arrhythmia.

4. Any infection requiring hospitalization or treatment with IV anti-infectives within 28 days of first dose of study drug.
5. Uncontrolled HIV, HBV, or HCV infection; or other uncontrolled infection or unexplained signs of infection (eg, fever of unknown origin, rising WBC count).
   a. Participants with HIV who have controlled infection (undetectable viral load and CD4 count above 350 cells/microliter either spontaneously or on a stable antiviral regimen) are permitted.
   b. Participants with HBV (HBsAg positive) who have controlled infection (serum HBV DNA PCR that is below the limit of detection AND receiving antiviral therapy for HBV) are permitted. Participants with occult HBV infection (negative HBsAg and positive HBV DNA) observed during the screening period should be evaluated by a specialist for start of anti-viral treatment and may enroll if HBV DNA becomes negative.
   c. Participants who are HCV antibody positive (HCV Ab+) with controlled infection (undetectable HCV RNA by PCR either spontaneously or in response to a successful prior course of anti-HCV therapy) are permitted.
6. History of severe allergic reaction attributed to compounds with a similar chemical or biologic composition as the study drug or excipient.
7. Known hypersensitivity to both allopurinol and rasburicase.
8. Any severe acute or chronic medical condition which could impair the ability of the participant to participate in the study or interfere with interpretation of study results (eg, systemic infection unless anti-infective therapy is employed), or participant unable to comply with the study procedures.
9. Participants who are committed to an institution by virtue of an order issued either by the judicial or the administrative authorities.
10. Diagnosis or treatment of another malignancy within 3 years of enrollment with the exception of completely resected basal cell carcinoma or squamous cell carcinoma of the skin, a definitively managed in situ malignancy, or low-risk prostate cancer after curative therapy. Ongoing hormonal therapy for a history of either prostate cancer or breast cancer is acceptable provided there has been no evidence of disease progression for at least 3 years prior to enrollment.
11. Prior exposure to approved or investigational treatments for SMM or MM (including but not limited to conventional chemotherapies, immunomodulatory drugs, proteasome inhibitors, anti-CD38 antibodies). Concurrent use of bisphosphonates or the RANKL inhibitor denosumab is not permitted; however, prior antiresorptives given for the treatment of osteoporosis are permissible.
12. Ongoing corticosteroid treatment with a dose >10 mg prednisone or equivalent per day at the time of screening.

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13. History of progressive multifocal leukoencephalopathy, neurodegenerative condition, CNS movement disorder, or seizure within 12 months prior to study enrollment.
14. Live or live attenuated vaccination within 28 days of planned start of study drug; COVID-19 vaccination within 1 week of planned start of study drug including anticipated completion of COVID-19 vaccination series within 1 week of study drug initiation.
15. History of allogeneic stem cell transplantation.

7.3. Premature Withdrawal from the Study

A participant has the right to withdraw from the study at any time, for any reason, and without repercussion.

The investigator and/or sponsor have the right to withdraw a participant from the study if it is no longer in the interest of the participant to continue in the study, or if the participant's continuation in the study places the scientific outcome of the study at risk (eg, if a participant does not or cannot follow study procedures). An excessive rate of withdrawals would render the study uninterpretable; therefore, unnecessary withdrawal of participants should be avoided.

Participants who are withdrawn prematurely from the study will be asked to complete the early termination visit, as described in Section 9.1.3.

Rules for discontinuation of study treatment (permanent or temporary) are discussed in Section 8.2.3.

7.4. Replacement of Participants

Participants prematurely withdrawn from the study/study drug can be replaced, if needed, to ensure an adequate number of evaluable participants. The Medical/Study Director, in cooperation with the Study Biostatistician, will decide whether to replace a withdrawn participant.

Participants enrolled and withdrawn from study any time prior to first dose of study drug will not be considered evaluable for assessment and will be replaced with another participant. All participants who have received a dose of study drug will be included in an as-treated analysis.

8. STUDY TREATMENTS

The study interventions administered in the study are outlined in Table 6 for IMP, Table 7 for premedication as AxMP, and Table 9 for rescue medication as AxMP.

8.1. Investigational and Reference Treatments

8.1.1. Linvoseltamab Investigational Treatment

Linvoseltamab for IV infusion will be a fixed dose and will not be dependent on participant weight or body surface area. A synonym for linvoseltamab is REGN5458. Diluent is unnecessary for most doses and may become unnecessary for all study doses (Table 6). See the pharmacy manual for details on whether diluent is employed for a given dose.

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A pharmacist or other qualified individual will be identified at each site to prepare linvoseltamab for administration. Instructions on dose preparation are provided in the pharmacy manual.

Table 6: Investigational Medicinal Product

Intervention Name	Linvoseltamab (REGN5458)	Diluent for Linvoseltamab
Dose Formulation	Sterile solution	Sterile solution
Unit Dose Strength(s)	10 mg/mL	N/A
Dosage Level(s)	See Section 6.1.1	N/A
Route of Administration	IV infusion	IV infusion
Use	Experimental	Diluent
Medicinal Product Type	IMP	IMP
Sourcing	Provided centrally by the sponsor	Provided centrally by the sponsor, if needed
Packaging and Labeling	Linvoseltamab will be provided in sterile single-use vials. Each vial will be labeled as required per country requirements.	Diluent for linvoseltamab will be provided in sterile single-use vials. Each vial will be labeled as required per country requirements.

IMP=investigational medicinal product; IV=intravenous

Each participant will start linvoseltamab with a step-up schedule: 1 mg at week 1, 4 mg at week 2, and 25 mg at week 3. The full dose of 200 mg will be given weekly for the first cycle followed by Q2W for the next 4 cycles. Starting at cycle 6, the dosing frequency decreases to Q4W. Treatment may continue for up to 24 cycles or until any criterion for study drug discontinuation is met.

The step-up doses and first full dose of linvoseltamab will be administered as continuous infusions over 4 hours. If the first full dose is adequately tolerated without CRS or IRR events of any grade, then subsequent linvoseltamab infusions may be reduced to 1 hour. If the 1-hour infusion time is tolerated, then subsequent infusions may be reduced to 30 minutes. See the pharmacy manual for details on preparation and administration.

Inpatient monitoring is required for the first 24 hours after each step-up dose. Additional inpatient monitoring can be implemented at the discretion of the investigator.

Note: The first 6 participants enrolled in the safety run-in cohort of the study who receive at least one 200 mg full dose and complete the 35-day safety observation period will be evaluated for tolerability. Based on the safety data reviewed after the safety run-in, the recommendations for inpatient monitoring may be updated.

8.1.2. Premedication

Premedication with dexamethasone, an antihistamine, and paracetamol (acetaminophen) is required prior to linvoseltamab infusions during the step-up period and first full dose according to Table 7.

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Table 7: Auxiliary Medicinal Product: Premedication for Linvoseltamab (Step-Up Period and First Full Dose)

Intervention Name	Dexamethasone	Antihistamine (Diphenhydramine or equivalent)	Paracetamol (Acetaminophen)
Sourcing	Site	Site	Site
Formulation	Determined by site	Determined by site	Determined by site
Unit Strength(s)			
Packaging and Labeling			
Dosage Level(s)	40 mg	25 mg	650 to 1000 mg
Route of Administration	IV (1 to 3 hours prior to dose)	IV or PO (30 to 60 minutes prior to dose)	PO (30 to 60 minutes prior to dose)
Use	Premedication	Premedication	Premedication
Medicinal Product Type	AxMP	AxMP	AxMP

AxMP=auxiliary medicinal product; IV=intravenous; PO=oral

If the participant experiences IRR and/or CRS of any grade upon administration of the first full dose of linvoseltamab, premedication should continue until the full dose is tolerated without signs of IRR and/or CRS.

If the full dose is adequately tolerated without CRS or IRR of any grade, the investigator may continue treatment without premedication or with a reduced dose of dexamethasone (eg, 10 mg).

If linvoseltamab is adequately tolerated with a tapered dose of dexamethasone, then discontinuation prior to subsequent treatment administration is encouraged.

When corticosteroids are discontinued, the antihistamine and acetaminophen may also be discontinued at the investigator's discretion.

8.2. Dose Modification and Study Treatment Discontinuation Rules

8.2.1. Linvoseltamab Dose Modification

Changes in dose (ie, dose reductions) are generally not allowed (see exceptions in Table 11). Delayed dosing of linvoseltamab is recommended in accordance with the below guidance or after consultation with the sponsor.

For clinically significant grade 3 hematologic toxicity:

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- Treatment should be paused in the following situations: grade 3 anemia, grade 3 thrombocytopenia with bleeding, grade 3 febrile neutropenia, or grade 3 neutropenia with documented infection
- Treatment (if paused) can be restarted when toxicity improves to grade ≤2 and, if applicable, bleeding or infection is adequately controlled

For clinically significant grade 4 hematologic toxicity:

- Treatment should be temporarily paused
- Treatment can be restarted when toxicity improves to grade ≤2
- Treatment should be permanently discontinued if toxicity does not resolve to grade <4 within 2 weeks of last infusion or to grade 2 within 8 weeks of last infusion with appropriate supportive measures

Participants who experience grade 3 non-hematologic toxicity not otherwise specified in this protocol should pause treatment until the toxicity improves to grade 1 or baseline. Exceptions to this guidance are permitted when laboratory abnormalities are considered clinically insignificant, or when health conditions unrelated to study treatment are stable and manageable with standard interventions. Study treatment should be permanently discontinued in the event of treatment-related grade 4 non-hematologic toxicity (see limited exceptions in Section 8.2.3.1).

For participants anticipated to restart therapy after a prolonged treatment interruption (>28 days), the sponsor should be consulted to discuss the participant's individual risk factors, such as prior history of CRS, before resuming treatment. See Table 8 for guidelines on treatment resumption after a dosing delay.

Table 8: Guidance on Treatment Resumption After Dosing Delay

Delay since last dose	Before full dose administration	After full dose administration
≤28 days	Repeat prior step-up dose or resume at the next planned step-up dose according to the treatment schedule.	Resume full dose according to the treatment schedule.
>28 days to ≤60 days	Permanently discontinue study treatment.	May resume full dose according to the treatment schedule.
>60 days	Permanently discontinue study treatment.	Permanently discontinue study treatment.

The investigator should carefully review with each participant the benefits and risks of restarting versus permanently discontinuing linvoseltamab after a dosing delay >28 days due to an AE. Participants who experience a treatment interruption of >60 days must discontinue treatment with linvoseltamab. When treatment is resumed >28 days after the administration of at least 1 full dose, premedications are recommended per Section 8.1.2. For shorter delays, premedications may be used at the investigator's discretion.

8.2.2. Dosing Schedule after Treatment Interruption

All participants must receive each of the incremental step-up doses during the step-up period before receiving the first full dose on C1D1. Repeat step-up doses are permitted during the step-up period to manage tolerability limited by CRS/IRR (Table 11).

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After full-dose therapy begins on C1D1, missed doses within an already started cycle will be skipped and not be made up; however, if a treatment interruption extends past the beginning of the next planned cycle, dosing will resume on the first scheduled dosing day of the next planned cycle. For example, if a participant has a 4-week delay after cycle 2 day 15, dosing will resume at the first scheduled dose in cycle 3 (C3D1).

8.2.3. Study Drug Discontinuation

Participants who permanently discontinue from study drug should be encouraged to remain in the study. Those who agree and do not withdraw from the study will be asked to return to the clinic for all remaining study visits per the follow-up schedule.

Participants who permanently discontinue from study drug and who opt to withdraw from the study will be asked to complete study assessments per Section 9.1.3.

8.2.3.1. Reasons for Permanent Discontinuation of Study Drug

Study drug dosing will be permanently stopped in the event of:

- Confirmed disease progression, as per IMWG response criteria (Appendix 1)
- Start of an alternative anti-cancer treatment
- Evidence of pregnancy
- Serious or severe allergic reactions considered related to study drug
- Specific types of liver dysfunction (eg, Hy's law) is met except in the context of CRS (FDA, 2009)
- Participant withdraws consent
- Any medical condition that the investigator or sponsor determines may jeopardize the participant's safety if the participant continues in the study or continues treatment with study drug
- The investigator determines it is in the best interest of the participant
- Participant non-compliance (eg, not complying with protocol-required visits, assessments, and dosing instructions)
- Tuberculosis or uncontrolled opportunistic infection
- Diagnosis of another malignancy during the study except non-melanoma skin cancer curable with local therapy or in situ cancers
- Any grade 4 toxicity

  Exceptions:

  - Isolated laboratory abnormalities without clinical consequence
  - Clinically significant laboratory abnormalities such as those associated with CRS that improve to grade <4 within 72 hours after onset
  - Hematologic laboratory abnormalities (Section 8.2.1)

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8.3. Safety Monitoring and Management of Specific Toxicities and Acute Reactions

Multiple measures will be taken to ensure the safety of the participants in this trial. This includes stringent eligibility criteria, close clinical safety monitoring according to the schedule of assessments, and requirement for hospitalization.

Upon occurrence of a toxicity, the investigator, with input from the sponsor, as needed, will assess the etiology of the toxicity. As a general principle, IRR and CRS commonly encountered during the step-up doses of linvoseltamab are more likely to warrant delays or dose modifications. Specific requirements on when to delay or modify the dose and the discontinuation criteria for linvoseltamab are provided in Section 8.2. Detailed guidance is provided below for specific AEs that are anticipated, such as acute allergic/hypersensitivity reactions, IRR/CRS, and ICANS. Guidance for the anticipated B-cell depletion and associated risk of infection as well as the potential need for TLS prophylaxis is also provided.

8.3.1. Acute Intravenous Infusion Reactions

Emergency equipment and medication for the treatment of infusion reactions must be available for immediate use. All infusion reactions must be reported as AEs (as defined in Section 10.2.4) and graded using the grading scales as instructed in Section 10.2.5.

8.3.1.1. Interruption of the Intravenous Infusion

The infusion should be interrupted if any of the following AEs are observed:

- Sustained/severe cough
- Rigors/chills
- Rash, pruritus (itching)
- Urticaria (hives, welts, wheals)
- Diaphoresis (sweating)
- Hypotension
- Dyspnea (shortness of breath)
- Vomiting
- Flushing

The reaction(s) should be treated symptomatically, and the infusion may be restarted at 50% of the original rate.

If investigators feel there is a medical need for treatment or discontinuation of the infusion other than described above, they should use clinical judgment to provide the appropriate response according to typical clinical practice.

8.3.1.2. Termination of the Intravenous Infusion

The linvoseltamab infusion should be terminated and NOT restarted if any of the following AEs occur:

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- Anaphylaxis*
- Laryngeal/pharyngeal edema
- Severe bronchospasm
- Chest pain
- Seizure
- Severe hypotension
- Other neurological symptoms (confusion, loss of consciousness, paresthesia, paralysis, etc.)
- Any other symptom or sign that, in the opinion of the investigator, warrants termination of the IV infusion

*Consider anaphylaxis if the following is observed (Sampson, 2006): acute onset of an illness (minutes to several hours) with involvement of the skin, mucosal tissue, or both (eg, generalized hives, pruritus or flushing, swollen lips-tongue-uvula) and AT LEAST ONE of the following:

- Respiratory compromise (eg, dyspnea, wheeze-bronchospasm, stridor, reduced peak expiratory flow, hypoxemia)
- Reduced BP or associated symptoms of end-organ dysfunction (eg, hypotonia [collapse], syncope, incontinence)

8.3.2. Grading and Management of Infusion-Related Reaction and Cytokine Release Syndrome

An acute IRR is defined as any AE that occurs less than 6 hours from the start of the infusion, or within 2 hours after completion of the infusion (whichever is later), and is associated with typical signs and symptoms including, but not limited to, flushing, tachycardia, hypotension, dyspnea, bronchospasm, back pain, fever, urticaria, edema, nausea, and rashes. Infusion-related reactions must be reported as AEs (Section 10.1.2) and graded according to the NCI-CTCAE v5.0 (Section 10.2.5).

CRS is a disorder characterized by fever, tachypnea, headache, tachycardia, hypotension, rash, and/or hypoxia; in this study, it is defined as such an event that occurs 6 or more hours from the start of the infusion or more than 2 hours after completion of the infusion (whichever is later). Emergency equipment and medication for the treatment of these potential adverse effects (eg, anti-cytokine therapy [such as tocilizumab], antihistamines, bronchodilators, IV saline, corticosteroids, acetaminophen, and/or epinephrine) must be available for immediate use (Table 9). Grade 2 or greater CRS must be reported as an AESI (Section 10.1.3) and graded according to the criteria adapted from (Lee, 2019) (Table 10).

Note: 6 hours as a time criterion to distinguish between CRS and other IRRs is established for reporting purposes only and is not a validated cutoff to distinguish between CRS and other types of IRRs. In any case, judgment of the investigator should be used, and if symptoms and/or laboratory evaluation suggest a CRS before 6 hours from the start of the infusion, the participant should be treated, and the event should be reported as CRS.

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IRRs and CRS have been observed in participants treated with linvoseltamab (see Section 3.2.4.2 and the linvoseltamab IB).

An extensive clinical and laboratory monitoring plan is in place to allow for early identification of participants who are more likely to develop CRS, and this allows for early initiation of increased participant monitoring and therapeutic intervention.

Participants who develop symptoms consistent with severe CRS, including but not limited to persistent fevers, neurologic disorders (including mental status changes, obtundation, and seizures), or clinical signs of toxicity (hypotension requiring at least 1 IV vasoactive pressor or hypoxia [pO2 <90%]), should be considered for pharmacologic intervention with anti-cytokine therapy (eg, tocilizumab and/or corticosteroids), and the sponsor should be contacted to discuss management (Davila, 2014) (Grupp, 2013). Additional guidance regarding the use of anti-cytokine therapy is provided in Table 12.

CRS and IRR management guidelines, including clinical evaluation and monitoring, are detailed in Table 10, Table 11, and Table 12.

Emergency equipment and medication for the treatment of infusion reactions must be available for immediate use.

Table 9: Auxiliary Medicinal Product: Rescue Medication

Intervention Name	Tocilizumab	Dexamethasone (Corticosteroid)	Others (Section 8.3.2)
Sourcing	Site	Site	Site
Formulation	Determined by site	Determined by site	Determined by site
Unit Strength(s)			
Packaging and Labeling			
Dosage Level(s)	4 to 8 mg/kg; dose not to exceed 800 mg	20 mg per day in divided doses	According to label
Route of Administration	Infusion	Infusion or oral	According to label
Use	Rescue medication	Rescue medication	Rescue medication
Medicinal Product Type	AxMP	AxMP	AxMP

AxMP=auxiliary medicinal product

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Table 10: Cytokine Release Syndrome Toxicity Grading

CRS Parameter	Grade 1	Grade 2	Grade 3	Grade 4
Fever[1]	Temperature ≥38˚C	Temperature ≥38˚C	Temperature ≥38˚C	Temperature ≥38˚C
With				
Hypotension	None	Not requiring vasopressors	Requiring vasopressors with or without vasopressin	Requiring multiple vasopressors (excluding vasopressin)
And/Or[2]				
Hypoxia	None	Requiring low-flow nasal cannula[3] or blow-by	Requiring high-flow nasal cannula[3], facemask, nonrebreather mask, or Venturi mask or blow-by	Requiring positive pressure (eg, CPAP, BiPAP, intubation and mechanical ventilation)

Adapted from ASTCT Cytokine Release Syndrome Consensus Grading (Lee, 2019).
Organ toxicities associated with CRS may be graded according to CTCAE v5.0, but they do not influence CRS grading.

[1] Fever is defined as temperature ≥38°C not attributable to any other cause. In participants who have CRS and then receive antipyretic or anti-cytokine therapy such as tocilizumab or steroids, fever is no longer required to grade subsequent CRS severity. In this case, CRS grading is driven by hypotension and/or hypoxia.

[2] CRS grade is determined by the more severe event: hypotension or hypoxia not attributable to any other cause. For example, a participant with temperature of 39.5°C, hypotension requiring 1 vasopressor, and hypoxia requiring low-flow nasal cannula is classified as grade 3 CRS.

[3] Low-flow nasal cannula is defined as oxygen delivered at ≤6 L/minute. Low flow also includes blow-by oxygen delivery, sometimes used in pediatrics. High-flow nasal cannula is defined as oxygen delivered at >6 L/minute.

BiPAP=bilevel positive airway pressure, CPAP=continuous positive airway pressure, CRS=cytokine release syndrome

Table 11: Management Guidance and Linvoseltamab Dosing in Setting of Cytokine-Mediated Toxicities & Infusion Related Reactions

Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
Acute IRR*	2	Temporarily pause study drug	Treat symptomatically: Supportive care, antipyretics (eg, paracetamol [acetaminophen]), meperidine.	No dose modification required. May resume when clinical symptoms are grade 1 or baseline. Consider an infusion rate of ≤50% of previous rate (up to 6 hr total). If the rate is reduced and tolerated, for subsequent infusions, the infusion rate may be increased to 100%.
	3	Temporarily pause study drug	Treat symptomatically: Supportive care, antipyretics (eg, paracetamol	May resume treatment when clinical symptoms are grade 1 or baseline.

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Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
			[acetaminophen]), meperidine. Consider corticosteroids. Vigilant supportive care is needed. The investigator should consider a possible CRS diagnosis and need for applicable CRS treatment.	Reduce dose by 50% from the previous dose with infusion rate of ≤50% of previous rate (up to 6 hr total). If the reduced dose is tolerated with no signs/symptoms of grade ≥3 IRR, the participant may return to the previous dose with ≤50% rate of infusion. The infusion rate may be increased to 100% if tolerable and, if tolerated, then proceed to the full dose. If grade 3 IRR recurs with subsequent doses, permanently discontinue treatment. Continue dosing if no recurrence of toxicity.
	4	Permanently discontinue study drug	Treat symptomatically: Supportive care, antipyretics (eg, paracetamol [acetaminophen]) and meperidine. Also consider corticosteroids. Vigilant supportive care is needed. The investigator should consider a possible CRS diagnosis and need for applicable CRS treatment.	Not applicable.
CRS*	1	Temporarily pause study drug	Vigilant supportive care (ie, assess for infection; treat febrile neutropenia if present, monitor fluid balance, antipyretics/ analgesics as needed); monitor cardiac and other organ functions closely; symptomatic management of organ toxicities. Initiate corticosteroids for participants with organ toxicities (listed in Table 11) excluding neurotoxicity. Consider corticosteroids and/or anti-cytokine therapy (eg, tocilizumab) as per institutional/consensus guidelines. Tocilizumab may be used for participants	No dose modification required. May resume when clinical symptoms are grade 1 or baseline. Consider decreasing the infusion rate to ≤50% of previous rate (up to 6 hr total). If the rate is reduced and tolerated, for subsequent infusions the infusion rate may be increased to 100%.

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Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
			with comorbidities, older age >65 years, or symptoms refractory to supportive measures (Reagan, 2021).	
	2	Temporarily pause study drug	Vigilant supportive care as per grade 1: Administer anti-cytokine therapy (eg, tocilizumab) as per institutional/consensus guidelines (Reagan, 2021). Initiate corticosteroids for participants with organ toxicities (listed in Table 9) excluding neurotoxicity.	No dose modification required. During the step-up dosing period, dose modification may be allowed after discussion with the sponsor including repeating or delaying a step-up dose. May resume when clinical symptoms are grade 1 or baseline. Consider hospitalization and consider decreasing the infusion rate to ≤50% of previous rate (up to 6 hr total). If the rate is reduced and tolerated, for subsequent infusions, the infusion rate may be increased to 100%.
	3	Temporarily pause study drug	Vigilant supportive care as per grade 2: Administer anti-cytokine therapy (eg, tocilizumab) and corticosteroids.	May resume linvoseltamab when symptoms of CRS are grade ≤1 or baseline. Hospitalize the participant and reduce dose by 50% from the previous dose with infusion rate of ≤50% of previous rate (up to 6 hr total). Consider prophylactic anti-cytokine therapy (eg, tocilizumab). If the reduced dose is tolerated with no signs/symptoms of grade ≥2 CRS, the participant may return to the previous dose with ≤50% rate of infusion. The infusion rate may be increased to 100% if tolerable. If the reduced dose of linvoseltamab is tolerated, then proceed to the full dose. If grade 3 CRS recurs with subsequent doses, permanently discontinue treatment.
	4	Permanently discontinue study drug	Vigilant supportive care as per grade 3.	Not applicable

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Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
Transaminase elevations (grade ≥3) associated with CRS*	≥3	Temporarily pause study drug	Supportive care as needed, monitor closely. Initiate corticosteroids immediately (see Table 12). Study drug will be temporarily withheld, and subsequent study drug treatment administrations will not resume until laboratory parameters are trending toward normal.	Resume linvoseltamab only when symptoms of CRS are grade ≤1 or baseline. No dose adjustment is required if the event is clinically uncomplicated and there is evidence of normalization within 7 days. Otherwise, reduce dose by 50%. Infusion rate adjustments should follow the guidance provided for each grade of CRS listed above.

If subsequent administration is tolerated, participants who received a reduced dose may have their dose increased at subsequent administration but no higher than the full dose, based on the clinical judgment of the investigator in consultation with the sponsor.
*CRS will be graded according to criteria specified in (Lee, 2019). All other events will be graded according to criteria specified in NCI-CTCAE v5.0. Consistent with these criteria, grade 1 infusion-related reactions require no infusion interruption or intervention.
CRS=cytokine release syndrome; IRR=infusion-related reaction

Table 12: Guidelines for Administration of Anti-Cytokine Therapy (eg, Tocilizumab) and Corticosteroids in Setting of Cytokine Release Syndrome

Event	Anti-Cytokine Therapy Dose
Left ventricular ejection fraction <40% by echocardiogram Creatinine >2.5-fold higher than the most recent level prior to linvoseltamab activated PTT >2 x ULN Clinically significant bleeding Creatine kinase >5 x ULN for longer than 2 days ALT/AST increase >5 x ULN	For example, 4 to 8 mg/kg tocilizumab infused over 1 hour, dose not to exceed 800 mg 20 mg dexamethasone per day in divided doses

Guidelines adapted from (Brudno, 2016). Note: This is not an exhaustive list.
ALT=alanine aminotransferase; AST=aspartate aminotransferase; PTT=partial thromboplastin time; ULN=upper limit of normal

8.3.3. ICANS

8.3.3.1. Study Definition of ICANS

ICANS is a stereotypic disorder characterized by a pathologic process involving the CNS following any immune therapy that results in the activation or engagement of endogenous or infused T cells and/or other immune effector cells. Symptoms or signs can be progressive and may include aphasia, altered level of consciousness, impairment of cognitive skills, motor weakness, seizures, and cerebral edema. ICANS may have features that overlap with other encephalopathies

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but has the more specific characteristic (when severe) of an awake patient who is mute and does not respond verbally or physically to an examiner (Lee, 2019).

Other neurological symptoms and/or signs may occur and possibly be attributable to immune effector cell-engaging therapies. As isolated signs/symptoms, these are not considered ICANS, and are considered under other neurologic events. For example, headache or tremor with no decrease in ICE score (Appendix 4) or no decreased level of consciousness, disorientation, or other stereotypic symptoms or signs of ICANS would not be considered ICANS. Neurologic symptoms may be observed in association with other pathological processes, including the examples listed below. Neurologic symptoms considered wholly explained by other processes like these would not be considered ICANS.

- Hepatic failure, severe hypertension, infection, electrolyte, or other laboratory abnormalities
- Adverse reactions to immunosuppressive therapies, cytotoxic drug therapies, or other concomitant medications
- Underlying disease processes (eg, a pre-existing seizure disorder)

The CARTOX-10-based assessment tool proposed by the ASTCT for grading neurotoxicity, termed the ICE score, is described in Section 9.2.3.4 and Appendix 4. The ICE score should be used to grade all cases of suspected ICANS in this trial.

8.3.3.2. Management of ICANS

When ICANS is suspected, participants should always undergo appropriate evaluation to rule out alternative etiologies (see Section 9.2.3.4 and Appendix 4). If a diagnosis of ICANS is made, site personnel must complete the dedicated ICANS case report form. Guidance regarding the management of ICANS events is provided in Table 13.

Table 13: ICANS Management Guidance

Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
ICANS	2	Refer to Section 6.1.3.3 and temporarily pause study drug	Treat symptomatically as per local guidelines. Suspend study drug at least temporarily until symptoms resolve to grade 1 or baseline. Participants should undergo appropriate evaluation to determine the etiology of the symptoms.	Any decision to restart treatment upon resolution of symptoms must be discussed and agreed upon by the investigator and the sponsor.
	≥3	Refer to Section 8.3.3 and discontinue if grade 4 or recurrent grade 3 ICANS	Treat symptomatically as per local or institutional guidelines. Participants should undergo appropriate evaluation to determine the etiology of the symptoms.	Participants who experience a single episode of grade 3 signs and symptoms of CNS toxicity (excluding headache) that is transient and resolves to grade 1 or baseline within 24 hours may continue the study drug regimen. Any decision to

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Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
				restart treatment after resolution of symptoms of a grade 3 event must be discussed and agreed upon by the investigator and the medical monitor. Participants who experience grade 4 or recurrent grade 3 toxicity consistent with ICANS must permanently discontinue study drug treatment.
Recurrent seizure	Any grade	Permanently discontinue study drug	Treat symptomatically as per local guidelines.	Not applicable.

CNS=central nervous system; ICANS=immune effector cell-associated neurotoxicity syndrome

8.3.4. B-Cell Depletion and Risk of Infection

Treatment with linvoseltamab may result in pronounced and prolonged B-cell depletion. The expected primary pharmacology of B-cell depletion may increase the risk of infections. In addition to B-cell depletion, high-risk SMM has been associated with immune system dysfunction, which may further contribute to immunodeficiency. Standard myeloma guidelines recommend increased vigilance and frequent reassessment of each individual participant for risk of infection as well as early signs of infection.

Participants with known active bacterial, viral, fungal, mycobacterial, or other infection, or any major episode of infection requiring hospitalization or treatment with IV anti-infectives within 4 weeks prior to first administration of linvoseltamab, are excluded from the trial. Infections of grade ≥3 will be reported as AESI (Section 10.1.3).

Recommendations for infection prevention:

- Vaccination against COVID-19, varicella zoster (recombinant vaccine), seasonal influenza and pneumococcus is strongly recommended for all participants, preferably prior to study enrollment. Local guidelines regarding COVID-19 boosters should also be followed.
- Antiviral prophylaxis with acyclovir or equivalent medication (per institutional standards) is recommended for all participants throughout the study duration to prevent herpes simplex virus and varicella zoster virus reactivation.
- PJP prophylaxis is required for all participants. Additional prophylaxis with antibacterial (eg, levofloxacin) and/or antifungal (eg, fluconazole) medication may be considered for high-risk participants according to local institutional guidelines.
- Monitoring for viral reactivation (eg, CMV, EBV, HHV-6, adenovirus) is recommended whenever new signs of infection or cytopenias are observed.
- Immunoglobulin replacement with IV immunoglobulin should be considered for participants with IgG concentrations less than 400 mg/dL.

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More comprehensive guidance on infection prevention is available from the IDSA, NCCN, and IMWG (Raje, 2022).

8.3.5. Tumor Lysis Syndrome

Tumor lysis syndrome is uncommonly encountered during the management of plasma cell dyscrasias. Experience with linvoseltamab among RRMM patients indicates that the risk of TLS is low, and study participants with SMM are considered less at risk than patients with MM. However, tumor lysis prophylaxis, such as with fluid hydration and allopurinol, may be considered on an individual basis per investigator discretion and institutional guidelines for participants who present with either an especially high tumor burden and/or indicators of rapid proliferation based on clinical (eg, high uric acid and/or lactate dehydrogenase), biological (eg, high Ki-67), or molecular criteria (eg, c-myc translocation).

8.4. Method of Treatment Assignment

All participants who sign the informed consent form (ICF) will be assigned a unique participant number by the Interactive Response Technology at screening. The assignment of treatment (linvoseltamab) will be performed in an unblinded fashion.

8.5. Blinding/Masking

This is an open-label study.

8.6. Treatment Logistics and Accountability

8.6.1. Packaging, Labeling, and Storage

Linvoseltamab (R5458, REGN5458) will be provided as vials within cartons. Vials and cartons will have country compliant labels. Open-label study drug will display the product lot number on the label.

Storage instructions will be provided in the pharmacy manual. Storage conditions of all drugs are provided on drug labeling and supersede all other sources of information.

8.6.2. Supply and Disposition of Treatments

Study drug will be shipped per labeled conditions to the investigator or designee at regular intervals or as needed during the study. At specified time points during the study (eg, interim site monitoring visits), at the site close-out visit, and following drug reconciliation and documentation by the site monitor, all opened and unopened study drug will be destroyed/returned to the sponsor or designee.

8.6.3. Treatment Accountability

All drug accountability records must be kept current.

The investigator must be able to account for all opened and unopened study drug. These records should contain the dates, quantity, and study medication:

- dispensed to each participant,
- returned from each participant (if applicable), and

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- disposed of at the site or returned to the sponsor or designee.

All accountability records must be made available for inspection by the sponsor and regulatory agency inspectors; photocopies must be provided to the sponsor at the conclusion of the study.

8.6.4. Treatment Compliance

All drug compliance records must be kept current and made available for inspection by the sponsor and regulatory agency inspectors.

8.7. Concomitant Medications and Procedures

Any treatment administered from the time consent is obtained until 90 days following the last dose of study drug or start of non-protocol cancer therapy, whichever occurs first, will be considered concomitant medication. This includes medications that were started before the study and are ongoing during the study.

All concomitant medications, including the generic name, dose, dose unit, frequency, start/stop date, and relative administration to linvoseltamab must be recorded in the study case report form (CRF). Any treatments that are started subsequent to the 90-day follow-up visit should not be recorded unless the medication is started to treat AEs deemed related to study drug.

8.7.1. Coronavirus Disease 2019 Vaccination During the Study

If a participant intends to receive a COVID-19 vaccine (initial series or booster) before the start of study drug administration, day 1 of the study and the events associated with day 1, should not occur until at least 1 week after any COVID-19 vaccination dosing.

COVID-19 vaccination dosing should not occur during the safety run-in observation period. If a participant who is enrolling intends to receive a COVID-19 vaccine within the safety run-in period, participation in the study should be delayed at least 1 week until after the final COVID-19 vaccine dose before starting linvoseltamab.

During the treatment period, it is recommended to delay COVID-19 vaccination, either as an initial series or as a booster dose, until participants are receiving and tolerating a steady dose of study drug.

COVID-19 vaccination, either as an initial series or as a booster dose received during the study, will be treated as a concomitant medication. As noted, administration of a COVID-19 vaccination should be separated from the time of administration of the investigational product (at least 72 hours, ideally by at least one week) to avoid confounding the effects (eg, adverse effects) of the vaccine/booster with the effects of study drug.

8.7.2. Prohibited Medications and Procedures

During the period defined in Section 8.7, unless otherwise specified in Section 8.7.3, a participant may not receive any of the following:

- Corticosteroid: Ongoing systemic treatment of more than 10 mg per day of prednisone or equivalent, except as specified in the protocol or for treatment of an AE
- Live or live attenuated vaccination with replicating potential

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- During the treatment period, any non-study treatment directed against plasma cells

8.7.3. Permitted Medications and Procedures

The following medications and procedures will be permitted, under the following conditions:

- Any medication required to treat an AE, including systemic corticosteroid therapy.
- Corticosteroid therapy for participants requiring physiologic replacement therapy, or requiring a brief course for prophylaxis (eg, contrast dye allergy), or for treatment of non-autoimmune conditions (eg, delayed-type hypersensitivity reaction caused by contact allergen).
- G-CSF, platelet, or red blood cell transfusions may be permitted for participants who require hematopoietic support.
- CYP450 substrates. Transient release of cytokines has been observed predominantly during initial weeks of dosing, and cytokines may suppress CYP450 enzymes (Morgan, 2001). For participants who are receiving concomitant CYP450 substrates, particularly those with a narrow therapeutic index, monitoring of the effect (eg, warfarin) or drug concentration (eg, cyclosporine or theophylline) is recommended, especially during the initial weeks of therapy.
- Any other medication which is considered necessary for the participant's welfare, and which is not expected to interfere with the evaluation of the study drug, may be given according to the physician's clinical judgment.

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9. STUDY SCHEDULE OF EVENTS AND PROCEDURES

9.1. Schedule of Events

Study assessments and procedures for the participants enrolled in Part 1 and Part 2 portions are presented by study period and visit in Table 14.

In the event of unforeseen circumstances (eg, COVID-19 or other national/international crisis), the continuity of clinical study conduct dictated by the schedule of events and oversight may require implementation of temporary or alternative mechanisms. Examples of such mechanisms may include, but are not limited to, any of the following: phone contact, virtual visits, telemedicine visits, online meetings, non-invasive remote monitoring devices, use of local clinic or laboratory locations, and home visits by skilled staff. Any study data collected using an alternative mechanism will be documented.

When multiple procedures are scheduled for the same visit/timepoint, the following guidance should be followed:

- PRO assessments should be completed before all other procedures and/or discussions about the participant's health.
- ECGs should be performed before blood draws, PET imaging, and study drug administration.
- All blood draws should be performed before ^{18}F-fluorodeoxyglucose (FDG) tracer administration as part of PET imaging and study drug administration. In addition:
  - PK/ADA samples should be obtained as soon as possible after an ECG
  - Safety and efficacy samples should be obtained before biomarker samples

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Table 14: Schedule of Events

	SCREENING	TREATMENT PERIOD													FOLLOW-UP PERIOD[1]		
		Step-Up Period			Cycle 1				Cycles 2-5				Cycles 6-24	EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
														Time After Last Linvoseltamab Dose			
Visit Week (W)		**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8 to W23**				**W24+**	**D30**	**W8 & W12**	**Q12W**	**Q24W**
Visit Day (D)	**-28 to -1**	**D1**	**D8**	**D15**	**D1**	**D8**	**D15**	**D22**	**D1**	**D8**	**D15**	**D22**	**D1**				
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±7	±30
Study Procedures[2]																	
Inpatient monitoring[3]		X	X	X													
Screening Only[4]																	
Informed consent	X																
Informed consent for optional FBR substudy[5]	X																
Inclusion/exclusion criteria	X																
Medical/oncology history	X																
Demographics	X																
Complete physical examination	X																
Height	X																
β2-microglobulin	X																
Echo or MUGA[2]	X																
HIV, HBV, HCV testing	X																
PT/INR and aPTT/PTT	X																

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	SCREENING	TREATMENT PERIOD													FOLLOW-UP PERIOD[1]		
		Step-Up Period			Cycle 1				Cycles 2-5				Cycles 6-24	EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
														Time After Last Linvoseltamab Dose			
Visit Week (W)		**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8 to W23**				**W24+**	**D30**	**W8 & W12**	**Q12W**	**Q24W**
Visit Day (D)	**-28 to -1**	**D1**	**D8**	**D15**	**D1**	**D8**	**D15**	**D22**	**D1**	**D8**	**D15**	**D22**	**D1**				
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±7	±30
Study Procedures[2]																	
Treatment																	
Premedication[6]		X	X	X	X												
Linvoseltamab[7]		X	X	X	X	X	X	X	X		X		X				
Efficacy/Disease Assessment																	
Overall response assessment					C1D1 then Q4W									X	X	X	
SPEP and SIFE[8,9]	X	X			C1D1 then Q4W									X	X	X	
24-hour urine for UPEP and UIFE[8,9,10]	X	X			C1D1 then Q4W									X	X	X	
Immunoglobulin quantification (IgA, IgM, IgG, IgD, IgE)[8,9]	X	X			C1D1 then Q4W									X	X	X	
IgG subclass testing IgG1, IgG2, IgG3, IgG4[8]		X			C1D1 then Q4W									X	X	X	
Serum FLC[9,10,11]	X	X			C1D1 then Q4W									X	X	X	
Bone marrow aspirate and/or core biopsy[2,12]	X				Time of ≥VGPR, 12 and 24 months after treatment initiation, then annually to confirm MRD, or at time of PD												
Whole body imaging (PET/CT and MRI)[2,13]	X				As clinically indicated												

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	SCREENING	TREATMENT PERIOD													FOLLOW-UP PERIOD[1]		
		Step-Up Period			Cycle 1				Cycles 2-5				Cycles 6-24	EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
														Time After Last Linvoseltamab Dose			
Visit Week (W)		**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8 to W23**				**W24+**	**D30**	**W8 & W12**	**Q12W**	**Q24W**
Visit Day (D)	**-28 to -1**	**D1**	**D8**	**D15**	**D1**	**D8**	**D15**	**D22**	**D1**	**D8**	**D15**	**D22**	**D1**				
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±7	±30
Study Procedures[2]																	
Long-term outcomes (eg, overall survival, progression to MM)																	X
Safety																	
Weight[2]	X	X			X				X				X	X			
Vital signs[14]	X	X	X	X	X				X				X	X	X		
Limited physical examination[2,15]		X	X	X	X	X	X	X	X		X		X	X	X		
Brief neurologic examination/ICE score[16]		X															
Electrocardiogram	X	As clinically indicated															
ECOG performance status[2]	X	X			X				X				X	X	X		
Adverse events		Continuous reporting														Only trSAE	
Concomitant medication		Continuous reporting														Only CMs for trSAE or anti-cancer therapy	

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	SCREENING	TREATMENT PERIOD													FOLLOW-UP PERIOD[1]		
		Step-Up Period			Cycle 1				Cycles 2-5				Cycles 6-24	EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
														Time After Last Linvoseltamab Dose			
Visit Week (W)		**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8 to W23**				**W24+**	**D30**	**W8 & W12**	**Q12W**	**Q24W**
Visit Day (D)	**-28 to -1**	**D1**	**D8**	**D15**	**D1**	**D8**	**D15**	**D22**	**D1**	**D8**	**D15**	**D22**	**D1**				
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±7	±30
Study Procedures[2]																	
Laboratory Testing (Local)																	
Hematology[2,17]	X	X	X	X	X	X	X	X	X		X		X	X	X		
CD4 and CD8 lymphocyte count[17]	X				C1D1, then Q12W for the first 12 months												
Blood chemistry[2,17]	X	X	X	X	X	X	X	X	X		X		X	X	X		
Urinalysis[17]	X																
Pregnancy test[2,18]	X (serum)				X				X				X				
CRP and ferritin		For CRP and ferritin sample collection time points, refer to Table 15															
Pharmacokinetics and Immunogenicity Samples																	
PK/drug conc. sample[19]		For PK/drug concentration sample collection time points, refer to Table 15															
ADA sample[20]		For ADA sample collection time points, refer to Table 15															
Biomarker Samples for Central Laboratory (Peripheral Blood)																	
Linvoseltamab interference serum sample[8]		X			C1D1, then Q4W whenever response assessment labs are drawn									X			
Immunophenotyping and RNA analysis[8]		X	X	X	X				C3D1, C4D1, then at suspected CR or PD					X			
Serum cytokine profiling		For serum cytokine profiling sample collection time points, refer to Table 15															

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	SCREENING	TREATMENT PERIOD													FOLLOW-UP PERIOD[1]		
		Step-Up Period			Cycle 1				Cycles 2-5				Cycles 6-24	EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
														Time After Last Linvoseltamab Dose			
Visit Week (W)		**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8 to W23**				**W24+**	**D30**	**W8 & W12**	**Q12W**	**Q24W**
Visit Day (D)	**-28 to -1**	**D1**	**D8**	**D15**	**D1**	**D8**	**D15**	**D22**	**D1**	**D8**	**D15**	**D22**	**D1**				
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±7	±30
Study Procedures[2]																	
Serum for exploratory biomarker[8]		X	X	X	X				C3D1, C4D1, then at suspected CR or PD					X			
Plasma for exploratory biomarker[8]		X	X	X	X				C3D1, C4D1, then at suspected CR or PD					X			
Serum for FBR use (optional)[5,8]		X	X	X	X				C3D1, C4D1, then at suspected CR or PD					X			
Plasma for FBR use (optional)[5,8]		X	X	X	X				C3D1, C4D1, then at suspected CR or PD					X			
DNA pharmacogenomics sample[8]		X												X			
MRD (blood)[8]	X				Time of ≥VGPR, 6, 12, 18, and 24 months after treatment initiation, then annually to confirm MRD												
Biomarker Samples for Central Laboratory (Bone Marrow Aspirate and Biopsy)[12]																	
MRD (BMA)[8]	X				Time of ≥VGPR, 12 and 24 months after treatment initiation, then annually to confirm MRD for up to 3 years after the last dose of linvoseltamab												
IHC/Spatial Transcriptomics	X																
Patient-Reported Outcomes Assessments																	
EORTC QLQ-C30[21]	X	X			X				X				X	X	X	X	
EORTC QLQ-MY20[21]	X	X			X				X				X	X	X	X	
PGIS[21]	X	X			X				X				X	X			

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	SCREENING	TREATMENT PERIOD													FOLLOW-UP PERIOD[1]		
		Step-Up Period			Cycle 1				Cycles 2-5				Cycles 6-24	EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
														Time After Last Linvoseltamab Dose			
Visit Week (W)		**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8 to W23**				**W24+**	**D30**	**W8 & W12**	**Q12W**	**Q24W**
Visit Day (D)	**-28 to -1**	**D1**	**D8**	**D15**	**D1**	**D8**	**D15**	**D22**	**D1**	**D8**	**D15**	**D22**	**D1**				
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±7	±30
Study Procedures[2]																	
PGIC[21]					X				X				X	X			
FACIT Item GP5[21]		X	X	X	X	X	X	X	X				X	X			

ADA=anti-drug antibody; aPTT=activated partial thromboplastin time; C1D1=cycle 1 day 1; CM=concomitant medication; CRP=C-reactive protein; CT=computed tomography; DNA=deoxyribonucleic acid; Echo=echocardiogram; ECOG=Eastern Cooperative Oncology Group; EMP=extra-medullary plasmacytoma; EORTC QLQC30=European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Core 30; EORTC QLQ-MY20= European Organization for Research and Treatment of Cancer Quality of Life Questionnaire-Multiple Myeloma Module 20; EOT=end of treatment; FACIT Item GP5=Functional Assessment of Chronic Illness Therapy Item GP5; FBR=future biomedical research; FLC=free light chain; HBV=hepatitis B virus; HCV=hepatitis C virus; HIV=human immunodeficiency virus; ICE=immune effector cell encephalopathy; Ig=immunoglobulin; IHC=immunohistochemistry; INR=international normalized ratio; MRD=minimal residual disease; MM=multiple myeloma; MRI=magnetic resonance imaging; MUGA=multi-gated acquisition scan; O=if at any time during the follow-up period assessment is performed because either clinically indicated, or as part of standard of care, collection of that information in the CRF is requested; PD=progressive disease; PET/CT=positron emission tomography/computed tomography; PK=pharmacokinetic; PT=prothrombin time; PTT=partial thromboplastin time; Q4W=every 4 weeks; Q12W=every 12 weeks; Q24W=every 24 weeks; Q26W=every 26 weeks; RNA=ribonucleic acid; SAE=serious adverse event; Seq=sequencing; SIFE=serum immunofixation; SPEP=serum protein electrophoresis; trSAE=treatment-related serious adverse event; UIFE=urine immunofixation; UPEP=urine protein electrophoresis; VGPR=very good partial response; W=week

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9.1.1. Footnotes for the Schedule of Events Table

1. End of Treatment (EOT): participants who discontinue study treatment as per Section 8.2.3, but who do not withdraw from the study, must return to the clinic for an EOT visit (Section 9.1.3). The follow-up period starts after the end of treatment visit and consists of 1) post-treatment follow-up where safety monitoring continues for up to 90 days and efficacy assessments continue until confirmed biochemical progression, start of non-protocol cancer therapy, or death; and 2) long-term follow-up that starts after the end of post-treatment follow-up to collect overall survival as well as other relevant long-term outcomes and continues for up to 5 years or until death or study termination by the sponsor, whichever occurs earlier.
2. Clinical assessments (ie, physical exam, weight, ECOG performance status, hematology, blood chemistry, and pregnancy test) required on week 1 day 1 (W1D1) prior to the first step-up dose of linvoseltamab may be performed up to 3 days before the visit, unless otherwise specified. Clinical assessments during treatment may be performed within 24 hours prior to study drug administration, unless otherwise specified. The following assessments do not need to be repeated if they have been performed within 8 weeks (56 days) prior to W1D1: whole body imaging, echocardiogram (Echo) or MUGA, or bone marrow biopsy. Submission of archival tissue from a biopsy performed before the study screening period is required. If unavailable, a repeat bone marrow biopsy must be performed.
3. All participants must be observed for at least 24 hours in a monitored inpatient setting after the end of infusion of each step-up dose of linvoseltamab.
4. For laboratory abnormalities or other clinical parameters that are clearly reversible, the screening window may be extended an additional 7 days to allow for correction and re-testing of the specific parameter(s) that do not meet entry criteria.
5. Informed Consent for Optional FBR Substudy: for participants who consent to separate optional Future Biomedical Research substudy.
6. Premedications include dexamethasone, an antihistamine, and paracetamol (acetaminophen). See Section 8.1.2 for further details.
7. Dosing days must be at least 5 days apart. C1D1 begins when the first full dose of linvoseltamab is administered after completion of the step-up schedule.
8. Samples will be collected predose and submitted to the central laboratory for analysis.
9. A second response assessment will be required to confirm response or progression. This can occur at a regularly scheduled visit using a separate response confirmation kit.
10. If the screening sample is collected within 14 days of W1D1, then there is no requirement to repeat collection of 24-hour urine before the W1D1 time point.
11. Free light chains may be tested as an exploratory biomarker measure using exploratory serum samples at additional time points other than those prespecified in Table 15.
12. A bone marrow aspirate and core biopsy sample will be collected for all participants at screening. The screening bone marrow aspirate sample will be used to assess chromosomal

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abnormalities, immunophenotypes by flow cytometry, baseline clonality for MRD testing, and other exploratory analyses. Subsequent bone marrow samples (aspirate and/or core biopsy) will be collected at the time when VGPR or better is achieved, 12 months after treatment initiation, 24 months after treatment initiation, and annually thereafter (if ongoing response). A bone marrow biopsy at the time of biochemical progression is required as well.

13. Local assessment for myeloma-defining radiologic findings will be completed at screening. See Section 9.2.2.2 for recommended imaging procedures.
14. Step-up doses (week 1 to week 3): vital signs will be taken pre dose/infusion, every 30 minutes (±15 mins) for the first 4 hours until EOI, and then hourly (±15 mins) until 4 hours following EOI and subsequently every 4 hours (± 1 hour) until 24 hours after EOI. From first full dose on C1D1 to end of therapy: vital signs will be taken pre-infusion, every 30 minutes (±15 mins) for the first 1 hour from start of infusion, then hourly (±15 mins) until EOI, as applicable. Vital signs on days without dosing may be taken at any time during the visit, and pulse oximetry is not required.
15. A complete physical examination may be completed, if clinically indicated.
16. A brief neurologic examination and ICE score assessment are required before administration of the first step-up dose of linvoseltamab on W1D1. Subsequent assessments should be performed as clinically indicated during treatment to monitor for neurotoxicity and/or ICANS (Appendix 4).
17. Hematology, blood chemistry, and urinalysis: see Section 9.2.3.6 for list of analytes.
18. Applicable to women of childbearing potential only; a serum β-hCG must be obtained during screening within 72 hours of first dose administration. After screening, subsequent tests may be performed on urine.
19. PK/Drug concentration sample: Soluble BCMA may be evaluated at time points when PK will be assessed (Table 15).
20. During the treatment period, ADA serum samples will be collected pre-dose on day 1 at the visits listed in Table 15. In the event of AESI, samples may be collected for ADA analysis at or near the event when possible.
21. All PRO questionnaires will be administered and completed during the scheduled visit prior to any other study procedures. The baseline EORTC QLQ-C30, EORTC QLQ-MY20, FACIT Item GP5, and PGIS assessments will be collected on C1D1 prior to the first dose of linvoseltamab. Where applicable, the questionnaires will be administered in the following order: EORTC QLQ-C30, EORTC QLQ-MY20, PGIS, PGIC, FACIT Item GP5.

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9.1.2. Detailed Sample Collection Time Points

Table 15: Detailed Sample Collection Time Points

Visit		Time Point	Serum Cytokines	PK	ADA	CRP and Ferritin
Screening		N/A				
Step-Up Doses	**Day 1 (Week 1)**	Predose	X	X	X	X
		EOI (T0)	X	X		
		T4H	X	X		X
	Day 8 (Week 2)	Predose	X	X		X
		EOI (T0)	X	X		
		T4H	X	X		X
	Day 15 (Week 3)	Predose	X	X		X
		EOI (T0)	X	X		
		T4H	X	X		X
Cycle 1	**Day 1**	Predose	X	X		X
		EOI (T0)	X	X		
	Day 8	Predose		X		X
		EOI (T0)	X	X		
	Day 15	Predose		X		X
		EOI (T0)	X	X		
	Day 22	Predose		X		X
		EOI (T0)		X		
Cycle 2	**Day 1**	Predose		X		
		EOI (T0)		X		
Cycle 3	**Day 1**	Predose		X	X	
		EOI (T0)		X		
Cycle 4	**Day 1**	Predose		X		
		EOI (T0)		X		
Cycle 5	**Day 1**	Predose		X		
		EOI (T0)		X		
Cycles 6-24	**Day 1**	Predose		Cycles 6, 12, 18, and 24	Cycles 6, 12, 18, and 24	
		EOI (T0)		Cycles 6, 12, 18, and 24		
30 Days After Last Dose		Any time during visit		X	X	

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Visit	Time Point	Serum Cytokines	PK	ADA	CRP and Ferritin
Week 8 and Week 12 After Last Dose	Any time during visit		X	Week 12 after last dose	

ADA=anti-drug antibody; CRP=C-reactive protein; D=day; EOI=end of infusion; H=hour; PK=pharmacokinetic; T=time

9.1.3. End of Treatment Visit

Participants who finish treatment or discontinue study treatment as per Section 8.2.3, but who do not withdraw from the study, must return to the clinic for an end of treatment visit (30 days [±7 days] after last dose or prior to the start of non-protocol cancer therapy, whichever occurs earlier) and will be expected to complete all applicable follow-up assessments (eg, efficacy and/or safety monitoring) as per the Schedule of Events (Table 14). If a participant is unable to return for an end of treatment visit, every effort should be made to contact the participant and collect adverse event information.

A participant who withdraws consent to continue participation in the study will not be followed for any reason after consent has been withdrawn.

9.2. Study Procedures

9.2.1. Procedures Performed Only at the Screening/Baseline Visit

The following procedures will be performed for the sole purpose of determining study eligibility or characterizing the baseline population as per Table 14:

- Medical/oncology history
- Demographics
- Complete physical examination
- Height
- β2-microglobulin
- Echocardiogram or MUGA (cardiac ejection fraction)
- HIV, HBV, HCV testing
- PT/INR and aPTT/PTT

If a participant has an ineligible test result during screening (eg, laboratory result), it is permissible to repeat the test during the screening period (see footnote #4 in Section 9.1.1 for details regarding the screening window). If multiple assessments are performed during screening, the most recent value will be used for eligibility purposes.

Rescreening is allowed once for participants who failed screening at an earlier time point and a new ICF will need to be signed at the re-screen visit. Some procedures may not need to be repeated if they were previously completed within 28 days (±7 days) prior to first dose.

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The participant's clinical laboratory assessments must be repeated prior to week 1 day 1 dosing if they were collected >3 days prior to dosing, and the participant must continue to meet all eligibility criteria prior to administering the first dose of linvoseltamab.

9.2.2. Efficacy Procedures

All efficacy procedures will be conducted at the time points indicated in the Schedule of Events (Table 14). Efficacy/disease assessments should be performed regularly at the prescribed interval (eg, every 4 weeks while on treatment) regardless of interruptions in treatment. If a participant is unable to undergo a disease assessment within the allowed window due to logistical or medical reasons, the disease assessment should be obtained at the next available date and before subsequent study treatment.

9.2.2.1. Overall Response Assessment

Disease assessments by investigators will be performed during the treatment and follow-up periods (ie, overall response assessment) according to IMWG criteria based on central laboratory analysis of bone marrow and peripheral blood samples. Characterization of serum and urine monoclonal protein will be performed by SPEP/SIFE and UPEP/UIFE, respectively. Serum FLCs and the corresponding ratio will be measured as well. Bone marrow core biopsy and aspirate will be assessed as described in Section 9.2.2.3 to enumerate plasma cell number at baseline and post-treatment. In addition, exploratory testing to evaluate any potential interference of linvoseltamab may be performed if an appropriate assay is available. This test may be performed on unused serum from standard MM disease assays or using exploratory research serum samples. Immunoglobulin quantification of various immunoglobulin subtypes (ie, IgG, IgA, IgM, IgD, IgE) and IgG subclass testing (Section 9.2.3.6) will also be performed at the time points indicated in the Schedule of Events (Table 14).

Two consecutive assessments will be required to confirm all responses. If a central laboratory result is not available, a result obtained locally and entered into the EDC may be used.

9.2.2.2. Baseline Whole Body Imaging

Whole-body imaging must be performed during screening for all participants to determine the presence of osteolytic skeletal lesions, intramedullary focal lesions, and/or plasmacytomas that would disqualify a participant from enrollment in the study (Hillengass, 2019). The preferred first imaging method is whole-body ^{18}F-FDG PET/CT, although whole-body low-dose CT is an acceptable alternative. If the CT component is negative, whole-body MRI (or MRI of the spine and pelvis if whole-body MRI is not available) must be performed to exclude focal lesions as myeloma-defining events. Image acquisition must conform to the quality standards specified in the imaging manual.

9.2.2.3. Bone Marrow Aspirate and/or Biopsy

All participants will require a bone marrow aspirate and core biopsy at screening to confirm a diagnosis of SMM. Subsequent bone marrow aspirates and/or biopsies will be collected according to the Schedule of Events (Table 14) and the below guidance:

- Upon confirmation of a response of VGPR or better, a bone marrow aspirate must be collected within 30 days to assess MRD status.

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- If biochemical progression occurs beyond 12 weeks from first administration of study drug, a bone marrow aspirate and/or core biopsy must be collected within 30 days.
- All participants who remain on study without evidence of biochemical progression must undergo bone marrow aspirate collection at 12 and 24 months for MRD assessment and evaluation of other molecular readouts.

Study-related bone marrow biopsy and/or aspirate samples must be shipped to central laboratories as described in the study laboratory manual. Evaluation methods may include but are not limited to immunohistochemistry, flow cytometry, FISH, MRD, and nucleotide sequencing (Section 9.2.7.4). Detailed instructions for sample collection and preparation are provided in the laboratory manual.

9.2.2.4. Minimal Residual Disease

MRD is a prognostic marker and its negative status (ie, MRD-) is associated with favorable outcome in patients with MM (Martinez-Lopez, 2014). MRD will be assessed according to IMWG guidelines by either flow cytometry (eg, EuroFLOW™) and/or NGS methods (clonoSEQ® assay cleared by FDA for MM only). Samples may be evaluated by either NGS or flow cytometry, and in some cases both methods may be used to compare concordance as neither method has been fully validated in this participant population. Next-generation sequencing will be performed longitudinally if a baseline sample establishing clonotypes is available for participants at the time of VGPR or better. If the baseline sample is not evaluable, then only flow cytometry may be used to evaluate MRD at suspected VGPR or better. Samples for both methods will be collected at screening, VGPR or better, and longitudinal time points to assess MRD. MRD at 10^{-5} sensitivity will be analyzed.

Sample collection time points are noted for bone marrow sample collection in the Schedule of Events (Table 14). Collection procedures for each method will be detailed in the laboratory manual.

9.2.2.5. Overall Survival Data Collection

Every effort will be made to collect survival data every 24 weeks (Q24W) for all participants the Schedule of Events (Table 14), including participants who withdraw from the study for any reason but have not withdrawn consent to collect survival information. If the death of a participant is not reported, the date of last participant contact in this study will be used for determination of the participant's last known date alive. If death is reported, efforts must be taken to determine the cause of death. If follow-up occurs remotely, the investigator may choose to collect survival data for all participants discontinued from their site on a fixed biannual basis for practical reasons.

9.2.3. Safety Procedures

All safety procedures, including ECOG performance status, will be conducted at the time points indicated in the Schedule of Events (Table 14). Monitoring for safety will be continuous.

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9.2.3.1. Body Weight

Body weight will be assessed using calibrated scales. Participants should void (empty bladder) prior to weight assessment. Participants should be wearing undergarments only and no shoes during weight assessments. Body weight will be recorded to the nearest 0.1 kg.

9.2.3.2. Vital Signs

Vital signs, including temperature, BP, pulse, pulse oximetry (on dosing days), and respiration rate will be collected at study visits indicated in the Schedule of Events (Table 14). Blood pressure should be measured after the participant has been resting quietly for at least 5 minutes. Blood pressure may be obtained from a seated or recumbent position and should be done consistently throughout the study. When scheduled at the same time as other procedures, vital signs should be measured prior to clinical laboratory assessments, PK, or exploratory sample collection.

9.2.3.3. Physical Examination

A physical examination will be conducted at study visits as indicated in the Schedule of Events (Table 14).

Care should be taken to examine and assess any abnormalities that may be present, as indicated by the participant's medical history. Screening requires a complete physical examination; a limited physical examination may be performed at subsequent visits.

9.2.3.4. Neurologic Examination

A brief neurologic examination including elements of the ICE score will be performed and recorded at baseline and as clinically indicated thereafter according to the Schedule of Events (Table 14) to screen and monitor participants for potential symptoms of neurotoxicity and ICANS.

The CARTOX-10-based assessment tool proposed by the ASTCT, termed the ICE score, grades neurotoxicity by evaluating alterations in speech, orientation, handwriting, and concentration that are highly suggestive of the encephalopathy observed in participants with ICANS (Lee, 2019). Symptoms of ICANS may manifest as delirium, encephalopathy, aphasia, lethargy, difficulty concentrating, agitation, seizures, motor weakness, depressed level of consciousness, and cerebral edema. Symptoms excluded because they are less specific and are usually managed symptomatically are headaches, tremor, myoclonus, asterixis, and hallucinations (Lee, 2019). See Appendix 4 for additional information about the scoring method.

9.2.3.5. Electrocardiogram

ECGs should be performed before blood is drawn during visits requiring blood draws. A standard 12-lead ECG will be performed at visits according to the Schedule of Events (Table 14). Heart rate will be recorded from the ventricular rate and the pulse rate, QRS, RR, and QT (identify QTcF) intervals will be recorded. The ECG strips or reports will be retained with the source.

9.2.3.6. Laboratory Testing

Hematology, CD4 and CD8 lymphocyte count, chemistry, urinalysis, CRP, ferritin, and pregnancy testing samples will be analyzed by a local laboratory. Detailed instructions for blood sample collection are in the laboratory manual provided to study sites.

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If a specific laboratory test is not routinely performed at a center, and instead, the center performs a different but equivalent test (eg, CO_2 instead of bicarbonate or urea instead of blood urea nitrogen), the equivalent test should be performed, and results collected. At centers where the bicarbonate (or equivalent) test is not performed as part of the routine chemistry panel, the test may be omitted.

Samples for laboratory testing will be collected at visits according to Table 14. Tests will include:

Blood Chemistry

Sodium	Total protein, serum	Total bilirubin
Potassium	Creatinine	Uric acid
Chloride	Blood urea nitrogen (BUN)	
Carbon dioxide	Aspartate aminotransferase (AST)	
Calcium	Alanine aminotransferase (ALT)	
Glucose	Alkaline phosphatase	
Albumin	Lactate dehydrogenase (LDH)	

Hematology

Hemoglobin	Differential:
Hematocrit	Neutrophils
Red blood cells (RBCs)	Lymphocytes
White blood cells (WBCs)	Monocytes
Platelet count	Basophils
	Eosinophils
	Plasma Cells (if applicable)

Urinalysis

Color	Glucose	RBC
Clarity	Blood	Hyaline and other casts
pH	Bilirubin	Bacteria
Specific gravity	Leukocyte esterase	Epithelial cells
Ketones	Nitrite	Crystals
Protein	WBC	Yeast

Other Laboratory Tests

CD4+ and CD8+ T-cell panel: CD4+ and CD8+ will be measured at screening and other time points as noted in the Schedule of Events (Table 14).

Pregnancy Test: Serum β-hCG will be measured at screening (≤72 hours prior to first dose of study drug administration) and urine β-hCG will be measured at all other time points for WOCBP.

CRP and Ferritin: These inflammatory markers will be measured at the time points specified in the Schedule of Events (Table 14).

Serum Calcium Corrected for Albumin (derived): Levels of unbound/free calcium (ie, biologically active calcium) will be calculated by adjusting for abnormal albumin levels ("corrected serum calcium").

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Abnormal Laboratory Values and Laboratory Adverse Events

All laboratory values must be reviewed by the investigator or authorized designee.

Significantly abnormal test results that occur after start of treatment must be repeated to confirm the nature and degree of the abnormality. When necessary, appropriate ancillary investigations should be initiated. If the abnormality fails to resolve or cannot be explained by events or conditions unrelated to the study medication or its administration, the Medical/Study Director must be consulted.

The clinical significance of an abnormal test value, within the context of the disease under study, must be determined by the investigator.

Criteria for reporting laboratory values as an AE are provided in Section 10.1.1.

9.2.4. Drug Concentration and Measurements

Blood samples for the assessment of linvoseltamab concentration in serum will be collected after the first dose and throughout the treatment period, as listed in Table 15. Actual sample collection times will be recorded. Detailed instructions for blood sample collection are provided in the laboratory manual. Soluble BCMA concentration may be measured from selected PK time points.

Any unused samples may be used for exploratory research including the possible evaluation of sBCMA (Section 9.2.7.3) and assay development and/or validation, biomarker research (Section 9.2.7), and/or FBR (Section 9.2.8).

9.2.5. Immunogenicity Measurements and Samples

Samples for ADA assessment will be collected at time points listed in Table 14 and Table 15.

Blood samples for ADA assessment in serum will be collected at time points in Table 15. Detailed instructions for blood sample collection are provided in the laboratory manual.

Any unused samples may also be used for exploratory research including assay development and/or validation, biomarker research (Section 9.2.7), and/or FBR (Section 9.2.8).

9.2.6. PRO Assessment Procedures

PRO measurements will be assessed according to the Schedule of Events (Table 14).

PRO measurements will include validated instruments EORTC QLQ-C30 (Aaronson, 1993) and EORTC QLQ-MY20 (Cocks, 2007) (Stead, 1999), commonly used in MM clinical trials. Two global anchors, the PGIC and PGIS, will be used to aid in the interpretation of EORTC QLQ-C30 and EORTC QLQ-MY20 results (Gary, 2020) (Guy, 1976). In addition, the FACIT Item GP5 will be used to assess overall impact of treatment side effects.

- The EORTC QLQ-C30 is a 30-item, generic questionnaire used to assess HRQoL in patients with cancer (Cocks, 2007) (Stead, 1999). It is one of the most frequently used generic measures of HRQoL in cancer clinical trials. The EORTC QLQ-C30 assesses HRQoL across multiple domains, including global health status, global quality of life, functioning (physical, role, emotional, cognitive, and social functioning), symptom scales (fatigue, nausea and vomiting, and pain), and single items (dyspnea, insomnia, constipation, diarrhea, financial impact, and appetite loss).

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- The EORTC QLQ-MY20 is a 20-item, disease-specific module which was developed to be used in conjunction with the EORTC QLQ-C30 to further assess HRQoL in patients with MM (Cocks, 2007) (Stead, 1999). The EORTC QLQ-MY20 assesses quality of life across multiple domains, including future perspectives, body image, MM-specific symptoms, and treatment side effects.
- The PGIC is a single item used to evaluate all aspects of patients' health and assesses if there has been an improvement or decline in clinical status. It can be adapted to assess the patient's impression of the change in a specific symptom (eg, pain) or category of symptoms (eg, disease-related symptoms, abdominal symptoms, etc.) (Hurst, 2004) (Guy, 1976). The change in symptoms is assessed on a 7-point scale (-3=very much worse, -2= moderately worse, -1= a little worse, 0=about the same, 1=a little better, 2=moderately better, 3=very much better).
- The PGIS is a single item used to evaluate patients' perception of disease severity. It can be adapted to assess the patient's perception of the severity of a specific symptom (eg, pain) or category of symptoms (eg, disease-related symptoms, abdominal symptoms) (Guy, 1976). The severity of the symptoms will be assessed on a 5-point scale (1=no symptoms, 2=mild symptoms, 3=moderate symptoms, 4=severe symptoms, 5=very severe symptoms).
- FACIT Item GP5 is a recommended item by FDA in its recent draft guidance for cancer trials to assess the overall impact of treatment side effect as a core PRO (FDA, 2021). It uses a single item "I am bothered by side effects of treatment" on a 5-point scale (0 = not at all, 1 = a little bit, 2 = somewhat, 3 = quite a bit, 4 = very much).

Non-English-speaking participants will complete the questionnaire using validated language translations developed and recommended for each instrument.

9.2.7. Pharmacodynamic and Exploratory Biomarker Procedures

In this study, biomarker assessments will be performed to explore the mechanism of action of linvoseltamab in high-risk SMM, and to evaluate the predictive value of specific biomarkers with respect to safety, drug exposure, and efficacy.

Biomarker samples will be collected at time points according to Table 14 and Table 15.

The analyses performed in the sections below may or may not be reported in the CSR. The collected samples will be stored for up to 15 years after the final date of the database lock. The samples may be utilized for exploratory research of SMM/MM, the mechanism of linvoseltamab action, and the biological systems targeted by linvoseltamab. After 15 years, any residual samples will be destroyed.

9.2.7.1. Biomarker Assessments

Biomarker samples will be collected at time points according to the Schedule of Events (Table 14). These samples are distinct from the future biomedical research samples as outlined in Section 9.2.8.

Progressive disease marker/biomarker measurements may be performed in blood and bone marrow cells, serum, plasma, and urine samples to determine effects of linvoseltamab on biomarkers of

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SMM or relevant physiological and pathogenic processes. Serum for additional testing will be collected for exploratory assays (such as mass spectrometry and Olink) to evaluate changes in proteins following treatment with linvoseltamab.

The biomarkers studied are believed to be relevant to the pathophysiology of SMM, target engagement, mechanism of action of linvoseltamab, and/or possible adverse events.

9.2.7.2. Peripheral Blood Biomarkers

Samples for peripheral blood (plasma, serum and blood for cell phenotyping, RNA and DNA analyses) biomarker assessments will be collected at time points listed in the Schedule of Events (Table 14).

Biomarker assays may include, but will not be limited to, the following:

- Immunophenotyping of T-cell and B-cell subsets and other cell populations (including monocytes and NK cells) in peripheral blood (by flow cytometry or similar methods) will be performed to assess baseline and post-treatment changes.
- Additional exploratory analyses requiring blood for DNA or RNA isolation may include but are not limited to WES, bulk/single cell RNA sequencing analysis, microsatellite instability, T-cell and B-cell clonality, and immune receptor repertoire sequencing.
- MRD monitoring to evaluate the level of tumor reduction by linvoseltamab in peripheral blood.

9.2.7.3. Serum Cytokines and Other Biomarkers

Serum samples will be collected for cytokine analysis according to Table 15.

Serum may also be used to evaluate additional biomarkers of interest relating to SMM and linvoseltamab's mechanism of action, including the evaluation of exploratory FLC testing. Cytokine changes triggered by binding of linvoseltamab to T cells may reflect T-cell activation. Understanding the magnitude and time course of pharmacodynamic changes in serum cytokines may inform dosing and help understand safety and efficacy.

9.2.7.4. Bone Marrow Exploratory Analyses

If sufficient bone marrow aspirate or core biopsy tissue is available after assessing tumor burden and confirming CR and MRD, samples may be stored for analysis of DNA, RNA, or protein to quantify the following:

- B-cell and T-cell immune receptor repertoires by sequencing
- Number and distribution of marrow-infiltrating lymphocytes (MILs) relative to malignant plasma cells such as CD8+ T cells, CD4+ T cells, T regulatory cells, and tissue permitting, other immune cell types including B cells, myeloid-derived cells, NK cells, etc
- Expression levels (messenger RNA and/or protein) including but not limited to checkpoint modulators (analysis may include nucleic sequencing and flow cytometry methods as well as additional methodologies as required)

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- Mutations in known oncogenes and potential tumor neoantigens
- Expression of other B-cell antigens to monitor resistance or response such as CD38 and SLAMF7
- MRD monitoring to evaluate the depth of tumor reduction by linvoseltamab in bone marrow tissue
- Any bone marrow core biopsy may be evaluated by immunohistochemistry/immunofluorescence techniques to better understand the tumor microenvironment composition.

Tissue permitting additional biomarkers may be measured such as, exome sequencing, single-cell RNA analysis, microsatellite instability, T cell and B cell clonality, and immune receptor repertoires.

9.2.8. Future Biomedical Research (Optional)

Participants who agree to participate in the FBR substudy will be required to consent to this optional substudy before samples are banked for FBR.

Additional peripheral blood samples will also be collected from participants who agree to participate in this optional substudy (see Table 14 for collection times).

Residual biomarker samples for study-related research, as well as unused PK and ADA samples, will be stored for up to 15 years after the final date of the database lock (or for a shorter time if required per regional laws and regulations). The samples may be utilized for FBR that may or may not be directly related to the study, including being used as reference samples and assay development or validation. The results of these FBR analyses will not be presented in the CSR.

9.2.9. Pharmacogenomic Analysis

Whole blood samples for DNA and RNA extraction will be collected from peripheral blood and bone marrow cells for pharmacogenomic analyses to understand the genetic determinants of efficacy and safety associated with the treatments in this study and the molecular basis of SMM and related diseases (see Table 14).

These samples will be single coded as defined by the ICH guideline E15. Samples will be stored for up to 15 years after the final date of the database lock. If there are specific site or country requirements involving the pharmacogenomic analyses which the sponsor is unable to comply with, samples will not be collected at those sites.

The purpose of the pharmacogenomic analyses is to identify genomic associations with clinical or biomarker response to linvoseltamab in SMM and possible AEs. In addition, associations between genomic variants and prognosis or progression of SMM to MM may also be studied. These data may be used or combined with data collected from other studies to identify and validate genomic markers related to the study drug, target pathway, or SMM and related diseases.

Analyses may include sequence determination or single nucleotide polymorphism studies of candidate genes and surrounding genomic regions. Other methods, including whole-exome sequencing, whole-genome sequencing, T-cell receptor sequencing, DNA copy number variation, and transcriptome sequencing (or other methods for quantitating RNA expression) may also be

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performed. The list of methods may be expanded to include novel methodology that may be developed during this study or sample storage period. Results from the genomic analyses will not be reported in the CSR.

9.2.10. Unscheduled Visits

All attempts should be made to keep participants on the study schedule. Unscheduled visits may be necessary for repeat testing following abnormal laboratory results, for management of ongoing AEs, or other safety-related reasons, as warranted.

Relevant assessments that occur during unscheduled visits should be reported in the respective CRF for the visit date.

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10. SAFETY EVALUATION AND REPORTING

10.1. Recording and Reporting Adverse Events

10.1.1. General Guidelines

The investigator must promptly record all clinical events occurring during the study data collection, per the Schedule of Events (Table 14). Medical conditions that existed or were diagnosed prior to the signing of the Informed Consent Form will be recorded as part of medical history. Abnormal laboratory values and vital signs observed at the time of informed consent should also be recorded as medical history. Any subsequent worsening (ie, any clinically significant change in frequency and/or intensity) of a pre-existing condition that is temporally associated with the use of the study drug should be recorded as an AE.

At each visit, the investigator will determine whether any AEs have occurred by evaluating the participant. Adverse events may be directly observed, reported spontaneously by the participant, or collected by questioning the participant at each study visit. Participants should be questioned in a general way, without asking about the occurrence of any specific symptoms. The investigator must assess all AEs to determine seriousness, severity, and causality, in accordance with the definitions in Section 10.2. The investigator's assessment must be clearly documented in the site's source documentation with the investigator's signature. The investigator should follow up on SAEs and AESI until they have resolved or are considered clinically stable; AEs should be followed until they are resolved or last study visit, whichever comes first.

Always report the diagnosis as the AE or SAE term. When a diagnosis is unavailable, report the primary sign or symptom as the AE or SAE term with additional details included in the narrative until the diagnosis becomes available. If the signs and symptoms are distinct and do not suggest a common diagnosis, report them as individual entries of AE or SAE.

Laboratory results, vital signs, and other diagnostic results or findings should be appraised by the investigator to determine their clinical significance. Isolated abnormal laboratory results, vital sign findings, or other diagnostic findings (ie, not part of a reported diagnosis) should be reported as AEs if they are symptomatic, lead to study drug discontinuation, dose reduction, require corrective treatment, or constitute an AE in the investigator's clinical judgment.

For events that are serious due to hospitalization, the reason for hospitalization must be reported as the SAE (diagnosis or symptom requiring hospitalization). A procedure is not an AE or SAE, but the reason for the procedure may be an AE or SAE. Pre-planned (prior to signing the Informed Consent Form) procedures, treatments requiring hospitalization for pre-existing conditions that do not worsen in severity, and admission for palliative or social care should not be reported as SAEs (see Section 10.2 for definitions).

For deaths, the underlying or immediate cause of death should always be reported as an SAE.

Diagnosis of malignancy will not be considered an AE or SAE if it is clearly consistent with the typical progression pattern of the underlying condition (including time course, affected organs, etc). Clinical symptoms of progression may be reported as AEs if the symptom cannot be determined as exclusively due to progression of the underlying malignant process or does not fit the expected pattern of progression for the condition under study. If there is any uncertainty about an AE being attributable to the development of malignancy, it should be reported as an AE or SAE.

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During the follow-up period, any SAE deemed treatment-related by an investigator that occurs after the safety monitoring window ends (90 days after last dose of study drug) and during post-treatment follow-up should also be reported.

All AEs, SAEs, AESI, and pregnancy reports are to be reported according to the procedures in Section 10.1.3.

10.1.2. Reporting Procedure

All SAEs and AESI must be reported to the sponsor (or designee) by completing the AE eCRF immediately, but no later than 24 hours of learning of the event. For SAEs and AESI, a detailed narrative summarizing the course of the event, including its evaluation, treatment, and outcome should be provided on the AE CRF. Specific or estimated dates of event onset, treatment, and resolution should be included, when available. Medical history, concomitant medications, and laboratory data that are relevant to the event should also be summarized in the narrative. For fatal events, the narrative should state whether an autopsy was or will be performed and include the results if available.

Information not available at the time of the initial report must be documented in a follow-up report. Source documents (including hospital or medical records, diagnostic reports, etc) will be summarized in the narrative on the AE CRF and retained at the study center and made available upon request.

Urgent safety queries must be followed up and addressed promptly. Follow-up information and response to non-urgent safety queries should be combined for reporting to provide the most complete data possible within each follow-up.

10.1.3. Events that Require Expedited Reporting to Sponsor

The following events also require reporting to the sponsor (or designee) within 24 hours of learning of the event:

- **SAEs**
- **AESI** (serious and nonserious) for this study include the following:
  - Grade 2 or greater CRS (Lee, 2019)
  - Grade 2 or greater ICANS
  - Grade 3 or higher TLS
  - Grade 3 or greater IRR
  - Grade 3 or greater allergic reaction
  - Grade 3 or greater infection
  - HBV reactivation

The following events also require reporting to the sponsor (or designee) within 24 hours of learning of the event:

- **Overdose of study drug:** Accidental or intentional overdose of at least 2 times the intended dose of linvoseltamab should be reported as an AE, even if not associated

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with signs or symptoms. If the overdose is associated with signs or symptoms, the signs or symptoms should be reported separately according to AE or SAE requirements.

- **Pregnancy:** Although pregnancy is not considered an AE, it is the responsibility of the investigator to report to the sponsor (or designee), within 24 hours of identification, any pregnancy occurring in a female participant or female partner of a male participant during the study or within 6 months of the last dose of study drug. Any complication of pregnancy affecting a female study participant or female partner of a male study participant, and/or fetus and/or newborn that meets the SAE criteria must be reported as an SAE. Outcomes for all pregnancies should be reported to the sponsor.

10.1.4. Other Adverse Events of Special Interest that do not Require Expedited Reporting to Sponsor

Although these AESI do not require expedited reporting to sponsor, the following events are of interest and will involve specific data collection to characterize better:

- Grade 1 CRS
- Grade 1 ICANS
- Grade 1 or 2 IRR

10.1.5. Abnormal Laboratory, Vital Signs, or Electrocardiogram Results

The criteria for determining whether an abnormal objective test finding should be reported as an AE include:

- The test result is associated with accompanying symptoms, and/or
- The test result requires additional diagnostic testing or medical/surgical intervention, and/or
- The test result leads to a change in dosing (outside of protocol-stipulated dose adjustments), discontinuation of study drug, significant additional concomitant drug treatment, or other therapy.

Contact the Medical/Study Director in the event the investigator feels that an abnormal test finding should be reported as an AE, although it does not meet any of the above criteria.

Repeating an abnormal test, in the absence of any of the above conditions, does not constitute an AE. Any abnormal test result that is determined to be an error does not require reporting as an AE.

Evaluation of severity of laboratory abnormalities will be assessed according to the scale outlined in Section 10.2.5.

10.1.6. Follow-Up

Information for any non-SAE that starts during study treatment or within 90 days after last dose of study treatment will be collected from the time of the event until resolution of the event, or until the participant's last study visit, whichever comes first.

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SAE information will be collected from the time of the event until the event is considered chronic and/or stable, or when the participant initiates a new anti-cancer treatment regimen, whichever comes first. Serious AEs thought to be related to study treatment may be followed longer.

10.2. Definitions

10.2.1. Adverse Event

An AE is any untoward medical occurrence in a participant administered a study drug which may or may not have a causal relationship with the study drug. Therefore, an AE is any unfavorable and unintended sign (including abnormal laboratory finding), symptom, or disease which is temporally associated with the use of a study drug, whether or not considered related to the study drug (ICH, 1994).

For studies with PROs, the PRO data are generally not reportable as individual AEs and thus will not be reported or reconciled as such.

10.2.2. Serious Adverse Event

An SAE is any untoward medical occurrence that at any dose:

- Results in **death** – includes all deaths, even those that appear to be completely unrelated to study drug (eg, a car accident in which a participant is a passenger).
- Is **life-threatening** – in the view of the investigator, the participant is at immediate risk of death at the time of the event. This does not include an AE that, had it occurred in a more severe form, might have caused death.
- Requires inpatient **hospitalization** or **prolongation of existing hospitalization**. Inpatient hospitalization is defined as a hospital admission (any duration), or an emergency room visit for longer than 24 hours. Prolongation of existing hospitalization is defined as a hospital stay that is longer than was originally anticipated for the event or is prolonged due to the development of a new AE as determined by the investigator or treating physician.
- Results in persistent or significant **disability/incapacity** (substantial disruption of one's ability to conduct normal life functions).
- Is a **congenital anomaly/birth defect**
- Is an **important medical event** - Important medical events may not be immediately life-threatening or result in death or hospitalization but may jeopardize the participant or may require intervention to prevent one of the other serious outcomes listed above (eg, intensive treatment in an emergency room or at home for allergic bronchospasm; blood dyscrasias or convulsions that do not result in hospitalization; or development of drug dependency or drug abuse).

Hospitalization or death due solely to manifestations consistent with typical progression of underlying malignancy will not be considered an SAE.

Criteria for reporting SAEs must be followed for these events.

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10.2.3. Adverse Events of Special Interest

An AESI (serious or non-serious) is one of scientific and medical interest specific to the sponsor's product or program, for which ongoing monitoring and rapid communication by the investigator to the sponsor can be appropriate. Such an event might warrant further investigation to characterize and understand it.

10.2.4. Infusion Reactions

Infusion reactions are defined in Section 8.3. All infusion reactions must be reported as AEs as described in Section 10.1.1. Depending on the nature of the event, rapid communication by the trial sponsor to other parties (eg, regulators) might also be warranted (Section 10.1.3).

10.2.5. Severity

The severity of AEs, infusion reactions, and test findings classified as AEs will be graded according to the NCI-CTCAE v5.0 grading system.

Adverse events not listed in the NCI-CTCAE v5.0 will be graded according to the scale provided in Table 16.

Table 16: Grading System for Adverse Events

Grade	Severity	Description
1	Mild	Asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated
2	Moderate	Minimal, local, or noninvasive intervention indicated, limiting age-appropriate instrumental ADL[a]
3	Severe	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL[b]
4	Life-Threatening	Life-threatening consequences: urgent intervention indicated
5	Death	Death related to AE

[a] Instrumental ADL refer to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.
[b] Self-care ADL refer to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden.
ADL=activities of daily living; AE=adverse event

10.2.6. Causality

The investigator must provide causality assessment as whether or not there is a reasonable possibility that the drug caused the AE based on evidence or facts, the investigator's clinical judgment, and the following definitions. The causality assessment must be made based on the available information and can be updated as new information becomes available.

The following factors should be considered when assessing causality:

- Temporal relationship: time to onset vs time drug was administered
- Nature of the reactions: immediate vs long-term

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- Clinical and pathological features of the events
- Existing information about the drug & same class of drugs
- Concomitant medications
- Underlying and concurrent illnesses
- Response to dechallenge (drug discontinuation) or dose reduction
- Response to rechallenge (re-introduction of the drug) or dose increase, when applicable
- Participant's medical and social history

Causality to the study drug (including study drug administration):

- Related:
  - The AE follows a reasonable temporal sequence from study drug administration and cannot be reasonably explained by the nature of the event, participant's clinical state (eg, disease under study, concurrent diseases, concomitant medications), or other external factors.

  or

  - The AE follows a reasonable temporal sequence from study drug administration and is a known reaction to the drug under study or its class of drugs or is predicted by known pharmacology.
- Not Related:
  - The AE does not follow a reasonable sequence from study drug administration or can be reasonably explained by the nature of the event, participant's clinical state (eg, disease under study, concurrent diseases, and concomitant medications) or other external factors.

Causality to the study conduct (protocol specified procedure):

- Related:
  - The AE follows a reasonable temporal sequence from a protocol specified procedure and cannot be reasonably explained by the nature of the event, participant's clinical state (eg, disease under study, concurrent diseases, concomitant medications), or other external factors.
- Not Related:
  - The AE does not follow a reasonable sequence from a protocol specified procedure or can be reasonably explained by the nature of the event, participant's clinical state (eg, disease under study, concurrent diseases, and concomitant medications) or other external factors.

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10.3. Safety Monitoring

The investigator will monitor the safety of the study participant at the investigator's site(s) as per the requirements of this protocol and consistent with current GCP. Any questions or concerns should be discussed with the sponsor in a timely fashion. The sponsor will monitor the safety data from across all study sites. The Medical/Study Director will have primary responsibility for the emerging safety profile of the compound, but will be supported by other departments (eg, Global Patient Safety; Biostatistics and Data Management). Safety monitoring will be performed on an ongoing basis (eg, individual review of SAEs) and on a periodic cumulative aggregate basis.

10.4. Notifying Health Authorities, Institutional Review Board /Ethics Committee, and Investigators

During the study, the sponsor and/or the contract research organization (CRO) will inform health authorities, Institutional Ethics Committees (IECs), and the participating investigators of any SUSARs (Suspected Unexpected Serious Adverse Reactions) occurring in other study centers or other studies of the active study drug (linvoseltamab), as appropriate per local reporting requirements. In addition, the sponsor and/or CRO will comply with any additional local safety reporting requirements.

Upon receipt of the sponsor's notification of a SUSAR that occurred with the study drug, the investigator will inform the Ethics Committee (EC) unless delegated to the sponsor.

Event expectedness for study drug (linvoseltamab) is assessed against the Reference Safety Information section of the IB that is effective for expedited safety reporting.

At the completion of the study, the sponsor will report all safety observations made during the conduct of the trial in the Clinical Study Report to health authorities and IECs as appropriate.

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11. STATISTICAL PLAN

The statistical plan may be revised during the study to accommodate amendments to the clinical study protocol if necessary. Additional details, summaries, data conventions, and plans for further statistical analyses will be provided in the Statistical Analysis Plan (SAP) document for the study. The SAP is expected to be finalized prior to any interim analysis (IA) of efficacy data and aligned with the plan described in the corresponding protocol section. In case of differences between the protocol and the SAP, the specifications and plans described in the SAP will take precedence.

11.1. Statistical Hypothesis

This is an exploratory study to evaluate the safety and preliminary clinical activity of linvoseltamab and therefore, no formal hypothesis testing is planned.

The primary endpoints for the expansion portion (Part 2) are:

- CR by investigator assessment
- MRD negativity with a minimum sensitivity of 10^{-5} at 12 and 24 months after the start of treatment

11.2. Justification of Sample Size

At least 6 participants will be enrolled during the safety run-in part of the study and 34 participants will be enrolled in the expansion part with the selected dosing regimen for expansion. These will be combined to reach a minimum total sample size of 40 participants treated with the selected dosing regimen and who have at least one post-baseline response assessment. With this sample size, we can estimate the ≥CR rate with reasonable precision and explore additional preliminary clinical activity, safety, PK properties, and biomarker changes associated with the dosing regimen.

The 2-sided, 95% binomial exact CI calculated with 40 participants for different observed rate of complete or better responses are presented in Table 17.

Table 17: The 95% Binomial Exact Confidence Intervals for Possible Observed Complete Response Rates with 40 Participants

Observed Number of ≥ CRs	≥ CR Rate	95% CI
12	30%	(16.6, 46.5)
14	35%	(20.6, 51.7)
16	40%	(24.9, 56.7)
18	45%	(29.3, 61.5)
20	50%	(33.8, 66.2)
22	55%	(38.5, 70.7)

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Observed Number of ≥ CRs	≥ CR Rate	95% CI
24	60%	(43.3, 75.1)
26	65%	(48.3, 79.4)

CI=confidence interval; CR=complete response

11.3. Analysis Sets

11.3.1. Efficacy Analysis Set

The FAS will include all participants who receive at least 1 dose (any amount) of study drug and have at least one post-baseline response assessment. Efficacy endpoints will be analyzed using the FAS.

11.3.2. Safety Analysis Set

The SAF will include all participants who receive any amount of study drug. Treatment compliance/administration and all clinical safety variables will be analyzed using the SAF. Participants will be classified according to the actual dosing regimen received.

11.3.3. Pharmacokinetic Analysis Sets

The PK analysis set is a subset of the FAS comprising participants who have at least 1 non-missing PK result following the first dose of study drug.

11.3.4. Immunogenicity Analysis Sets

The immunogenicity analysis set is a subset of the FAS comprising participants who have at least 1 non-missing ADA result following the first study dose.

11.3.5. Soluble BCMA Analysis Sets

The sBCMA analysis set is a subset of the FAS comprising participants who have an available baseline and at least 1 post-dose sample collected.

11.3.6. Patient-Reported Outcome Analysis Sets

The PRO analysis set is a subset of the FAS comprising participants who have at least 1 non-missing PRO result following the first study dose.

11.3.7. Biomarker Analysis Set

The biomarker analysis set is a subset of the FAS comprising participants who have an available baseline biomarker result.

11.4. Statistical Methods

For continuous variables, descriptive statistics will include the following information: the number of participants reflected in the calculation (n), mean, standard deviation, Q1, median, Q3, minimum, and maximum.

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For categorical or ordinal data, frequencies and percentages will be displayed for each category.

11.4.1. Participant Disposition

The number and percentage of participants screened, randomized, and the primary reasons for screening failure and discontinuation will be summarized.

11.4.2. Demography and Baseline Characteristics

Demographic variables (eg, age, race, and gender), baseline characteristics, medical history, and prior and concomitant mediations and therapies will be summarized by treatment group. See Section 5.1 for a full list of demographic and baseline variables. No statistical hypothesis tests will be performed on these characteristics.

11.4.3. Efficacy Analyses

11.4.3.1. Primary Efficacy Analysis

The primary endpoints for the expansion portion (Part 2) are incidence of CR or better and MRD negativity determined by IMWG response criteria (Appendix 1).

The rate of MRD-negativity is defined as the proportion of participants who achieve MRD-negativity at 10^{-5}, among participants with ≥VGPR based on FAS. MRD negative rate at 12 months and at 24 months of treatment initiation will be summarized. For the primary analysis, participants with missing data will be treated as non-responders and will be counted in the denominator, but not in the numerator. The point estimate and exact 95% CI will be provided using the Clopper-Pearson method.

The ≥CR rate is defined as the proportion of participants with a best response of CR or better by IMWG response criteria (Kumar, 2016). For the primary analysis, participants with inevaluable responses will be considered as non-responders. The point estimate and exact 95% CI will be provided using the Clopper-Pearson method.

11.4.4. Secondary Efficacy Analysis

ORR is defined as the proportion of treated participants who achieve a best overall response of PR or better. The ORR and the corresponding 95% CI will be calculated using the Clopper-Pearson method.

DOR is defined for responders (participants who achieve PR or better) only. It is the time from the date of the first documented response (at least PR) until the date of first documented disease progression, or death due to any cause, whichever occurs first. For participants who are alive and have not experienced disease progression at the time of analysis, DOR will be censored at their latest available adequate disease assessments.

PFS is defined as the time from first dose date until the first date of PD, or death due to any cause, whichever occurs first. For participants who are alive and have not experienced disease progression at the time of analysis, PFS will be censored at their latest available adequate disease assessments.

Time to new anti-myeloma therapy (ie, new systemic therapy for MM) is measured as the time from the start of the study treatment to the initiation of systemic therapy for MM, or death due to

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any cause, whichever occurs first. If a participant is not known to have died or have received new systemic therapy for MM at the time of analysis, this endpoint will be censored at the last date the participant is documented to be receiving study treatment or to be free of new anti-myeloma therapy.

OS is measured as the time from the start of study treatment until death due to any cause. If a participant is not known to have died at the time of analysis , OS will be censored at the last date that the participant is documented to be alive.

The Kaplan-Meier curve and estimated median (or an appropriate percentile if the median is not reached) along with its 95% CI will be provided for each of the secondary time-to-event endpoints described above.

TTR is defined for responders (participants that achieve PR or better) only. It is the time from first dose date to the date of first response. Since this endpoint is not censored, descriptive statistics will be presented for TTR.

11.4.5. Control of Multiplicity

Not applicable since there is no formal hypothesis test.

11.4.6. Safety Analyses

11.4.6.1. Adverse Events

Definitions

Treatment Emergent Adverse Events (TEAEs) are defined as those AEs with either initial onset after the first dose of study treatment or that represent an exacerbation of a pre-existing condition after the first dose of study treatment.

Analysis

Summaries of TEAEs will include events that occur up to 30 days after last treatment administration or the day prior to the start of non-protocol cancer therapy, whichever comes first, as well as all treatment-related TEAEs regardless of timing.

The number (n) and percentage (%) of participants who reported any of the following will be summarized by treatment group:

- TEAEs by system organ class (SOC) and preferred term (PT)
- TEAEs by severity (according to the grading scale outlined in Section 10.2.5), SOC and PT
- Treatment-emergent AESI
- TEAEs leading to permanent treatment discontinuation
- SAEs by SOC and PT

11.4.6.2. Other Safety

Vital Signs

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Vital signs (temperature, pulse rate, pulse oximetry [on dosing dates], blood pressure and respiration rate) at baseline and change from baseline to each scheduled assessment time will be summarized with descriptive statistics.

Laboratory Tests

Laboratory test results at baseline and change from baseline at each scheduled assessment time will be summarized with descriptive statistics.

Number and percentage of participants with a PCSV will be tabulated.

11.4.6.3. Treatment Exposure

Treatment duration and relative dose intensity will be summarized.

11.4.6.4. Treatment Compliance

Not applicable.

11.4.7. Pharmacokinetics

11.4.7.1. Analysis of Drug Concentration Data

The PK parameters may include, but are not limited to:

- C_{max}
- C_{EOI}
- C_{trough}

The concentrations of linvoseltamab in serum over time and selected pharmacokinetic parameters will be summarized using descriptive statistics for the purpose of estimating PK exposures in participants with SMM. This descriptive statistical assessment will include the geometric means and ratios of the geometric means for selected PK parameters, as deemed appropriate.

No formal statistical hypothesis testing will be performed.

11.4.8. Analysis of Immunogenicity Data

The immunogenicity variables will be summarized using descriptive statistics. Immunogenicity will be characterized by ADA status, ADA category, and maximum titer observed in participants in the ADA analysis set. For samples confirmed as drug-specific ADA positive, but found negative at the lowest titer dilution, the lowest dilution in the titer assay is imputed.

The ADA status of each participant may be classified as one of the following:

- Positive
- Pre-existing - If the baseline sample is positive and all post-baseline ADA titers are reported as less than 9-fold the baseline titer value
- Negative - If all samples are found to be negative in the ADA assay

The ADA category of each positive participant is classified as:

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- Treatment-boosted - A positive result at baseline in the ADA assay with at least 1 post-baseline titer result 9-fold the baseline titer value
- Treatment-emergent - A negative result or missing result at baseline with at least 1 positive post-baseline result in the ADA assay. Participants that are treatment-emergent will be further sub-categorized as follows:
  - Persistent – A positive result in the ADA assay detected in at least 2 consecutive post-baseline samples, separated by at least a 16-week post-baseline period (based on normal sampling time), with no ADA-negative results in between, regardless of any missing samples
  - Transient – Not persistent or indeterminate, regardless of any missing samples
  - Indeterminate – A positive result in the ADA assay at the last collection time point only, regardless of any missing samples

The maximum titer category of each participant is classified as:

- Low (titer <1,000)
- Moderate (1,000≤ titer ≤10,000)
- High (titer >10,000)

The following will be summarized by treatment group and ADA titer level:

- Number (n) and percent (%) of ADA-negative participants
- Number (n) and percent (%) of pre-existing participants
- Number (n) and percent (%) of treatment-emergent ADA positive participants
  - Number (n) and percent (%) of persistent treatment-emergent ADA positive participants
  - Number (n) and percent (%) if indeterminate treatment-emergent ADA positive participants
  - Number (n) and percent (%) of transient treatment-emergent ADA positive participants
  - Number (n) and percent (%) of treatment-boosted ADA positive participants

Listing of all ADA titer levels will be provided for patients with pre-existing, treatment-emergent and treatment-boosted ADA response.

11.4.8.1. Immunogenicity with Exposure

Potential association between immunogenicity and systemic exposure to REGN5458 will be explored by treatment groups. Plots of individual REGN5458 concentration time profiles may be provided to examine the potential impact of ADA category and maximum titer category on these profiles.

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11.4.8.2. Immunogenicity and Safety and Efficacy

Potential association between immunogenicity variables and safety may be explored with a primary focus on the following safety events during the TEAE period:

- Infusion reactions
- Hypersensitivity (SMQ: Hypersensitivity [Narrow])
- Anaphylaxis (SMQ: Anaphylactic Reaction [Narrow])

Potential association between immunogenicity variables and efficacy endpoints may be explored (eg, scatter plot or spaghetti plot).

The safety and efficacy analyses mentioned above will be conducted using the following categories:

- ADA Positive
  - Treatment-emergent
  - Treatment-boosted

11.4.9. Maximum Post-Baseline Titer Category Analysis of Pharmacodynamic and Exploratory Biomarker Data

The following variables will be explored for the relationship with IMWG response:

- Analysis of pharmacodynamic variables, including the change in peripheral blood major T-cell subsets (CD3, CD4, CD8) over time.
- Baseline BCMA expression detected by immunohistochemistry on the bone marrow core biopsy will be analyzed descriptively as a continuous variable using intensity of staining in a percentage of cells as well as using the H score of cytoplasmic and membrane staining.
- Analysis of sBCMA in serum at baseline (by medians, quantiles such as tertiles) will be analyzed descriptively as subgroups relating to IMWG response. The concentrations of sBCMA in serum over time will be summarized by descriptive statistics.
- Analysis of baseline bone marrow T-cell percentages and major subsets (CD3, CD4, CD8) will be described.
- Additional exploratory biomarkers such as changes in circulating B cells, monocytes, and NK cells may be described in a separate report.

11.4.10. Analysis of Patient-Reported Outcome Data

For both cohorts, PRO data, as measured by validated instruments EORTC QLQ-MY20, EORTC QLQ-C30, 2 global anchors PGIS and PGIC, and FACIT Item GP5 will be described, including mean scores at each assessment and change from baseline scores over time.

Additional analyses of PRO data will be performed. Time to deterioration in global health status/QoL (per EORTC QLQ-C30) will be summarized using the Kaplan-Meier method and will include the median and 95% CI. Longitudinal mixed effects model analyses will be used to assess

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change from baseline in global health status/QoL. The model will include timepoint as a fixed-effect categorical factor, the baseline PRO score as a fixed-effect covariate, and the baseline vs time interaction. Evaluation on general health status, patient-reported function, and symptoms (per EORTC QLQ-C30 and QLQ-MY20) regarding time to deterioration and change from baseline will be summarized using methods mentioned above. Further details regarding these analyses will be included in the SAP. Missing items will be handled per the scoring manuals of each questionnaire administered.

PRO instruments will be scored according to their respective scoring manuals and user's guides. Patient disposition and PRO completion rates will be summarized by visit.

Summary statistics of absolute scores and change from baseline by visit will be reported for each scale/subscale of all PRO instruments. Line charts depicting the mean and mean change from baseline will be provided.

11.5. Interim Analysis

Two interim assessments of futility based on observation of objective responses and complete responses are planned in this study (see Section 6.2 for details).

- The first interim assessment will be performed after 15 participants have started treatment with linvoseltamab and completed at least 15 weeks of follow-up with at least one post-baseline response assessment. If 5 or fewer responses are observed or there are no CR observed, additional participants will not be enrolled.
- The second interim assessment will be performed after 30 participants have started treatment with linvoseltamab and completed at least 15 weeks of follow-up with at least one post-baseline response assessment. If 13 or fewer responses are observed or there are fewer than 3 CR observed, additional participants will not be enrolled. If 14 or more responses and 3 or more ≥CR are observed, the study will continue to full enrollment.

The OCs of the ORR-based decision rules under various possible ORR are summarized in Table 18. To avoid making additional assumptions, the CR-based conditions have been ignored in the calculation of these OCs. It is expected that no fewer than 15% of responses should be a CR. If the true ORR is as low as 30%, there is a 72% probability of stopping further enrollment after the first IA, and only a 4% chance of continuing to full enrollment after the second IA. On the other hand, if the true ORR is as high as 55%, then there is only an 8% probability of stopping further enrollment after the first IA, and an 84% chance of continuing to full enrollment after the second IA.

Table 18: Operating Characteristics of Futility Criteria

True ORR	Probability of stopping at the first interim look (N=15)	Probability of stopping at either first or second interim look (N=30)	Probability of continuing to full enrolment (N=30)
60%	0.03	0.07	0.93
55%	0.08	0.16	0.84

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50%	0.15	0.33	0.67
45%	0.26	0.53	0.47
40%	0.40	0.74	0.26
35%	0.56	0.89	0.11
30%	0.72	0.96	0.04
20%	0.94	0.99	0.01

ORR=overall response rate

No multiplicity adjustment is necessary at interim looks since no formal hypothesis testing is planned.

11.6. Statistical Considerations Surrounding the Premature Termination of a Study

If the study is terminated prematurely, only those parameters required for the development program and/or reporting to regulatory authorities will be summarized. Investigator and sponsor responsibilities surrounding the premature termination of a study are presented in Section 15.1.

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12. QUALITY CONTROL AND QUALITY ASSURANCE

In accordance with ICH E6, the sponsor is responsible for quality assurance to ensure that the study is conducted, and the data generated, recorded, and reported in compliance with the protocol, GCP, and any applicable regulatory requirement(s). The planned quality assurance and quality control procedures for the study are described in this section.

12.1. Data Management and Electronic Systems

12.1.1. Data Management

A data management plan specifying all relevant aspects of data processing for the study (including data validation [quality-checking], cleaning, correcting, releasing) will be maintained and stored at Regeneron (sponsor).

Adverse events and medical history will be coded using MedDRA. A medical coding plan will specify the processes and the dictionary used for coding. All data coding (eg, AEs, baseline findings, medication, medical history/surgical history) will be done using internationally recognized and accepted dictionaries.

The CRF data for this study will be collected with an EDC tool.

12.1.2. Electronic Systems

Electronic systems that may be used to process and/or collect data in this study will include the following:

- Interactive Response Technology participant assignment, study drug supply
- EDC system – data capture – Medidata Rave
- SAS – statistical review and analysis
- Pharmacovigilance safety database

12.2. Study Monitoring

12.2.1. Monitoring of Study Sites

Regeneron uses a study-specific, risk-based approach to study monitoring and oversight, aligned with risk-based quality principles, outlined in ICH E6 (R2) Guideline for Good Clinical Practice. RBQM methodology focuses on employing a fit-for-purpose monitoring strategy, supported either directly by Regeneron as sponsor, or via our CRO partners. RBQM strategies may include reduced SDV, targeted SDR, the use of onsite and remote monitoring visits (as allowed by local laws), and Centralized Monitoring to identify site level risks and study level trends. The investigator must allow study-related monitoring activities to occur.

The study monitors may perform targeted SDR to verify that data recorded in the CRF by authorized site personnel are accurate and verifiable from source documents, that the safety and rights of participants are being protected, and that the study is being conducted in accordance with the current approved protocol version and any other study agreements, ICH GCP, and all applicable regulatory requirements.

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12.2.2. Source Document Requirements

Investigators are required to prepare and maintain adequate and accurate participant records (source documents). The site is responsible to ensure quality within their records and systems and are accountable for ensuring that all source data and CRF data are timely, accurate and complete.

The investigator must keep all source documents on file with the CRF (throughout this protocol, CRF refers to either a paper CRF or an eCRF). Case report forms and source documents must be always available for inspection by authorized representatives of the sponsor and regulatory authorities.

12.2.3. Case Report Form Requirements

Study data obtained during the clinical study will be recorded on eCRFs within the EDC system by trained site personnel. All required CRFs must be completed for each participant enrolled in the study. The investigator must ensure the accuracy, completeness, and timeliness of the data reported to the sponsor in the CRFs. After review of the clinical data for each participant, the investigator must provide an electronic signature. A copy of each participant CRF casebook is to be retained by the investigator as part of the study record and must be always available for inspection by authorized representatives of the sponsor and regulatory authorities.

A paper CRF for each participant enrolled in the study must be completed and signed by the study investigator or authorized designee. The CRF must be typed or filled out using indelible ink. The writing must be legible. Errors should be crossed out but not obliterated, the correction inserted, and the change initialed and dated by the investigator or authorized designee. The investigator must ensure the accuracy, completeness, legibility, and timeliness of the data reported to the sponsor in the CRFs. CRFs must be available always for inspection by authorized representatives of the sponsor and regulatory authorities.

Corrections to the CRF will be entered in the CRF by the investigator or an authorized designee. All changes, including date and person performing corrections, will be available via the audit trail, which is part of the EDC system. For corrections made via data queries, a reason for any alteration must be provided.

12.3. Audits and Inspections

This study may be subject to a quality assurance audit or inspection by the sponsor or regulatory authorities. Should this occur, the investigator is responsible for:

- Informing the sponsor of a planned inspection by the authorities as soon as notification is received, and authorizing the sponsor’s participation in the inspection
- Providing access to all necessary facilities, study data, and documents for the inspection or audit
- Communicating any information arising from inspection by the regulatory authorities to the sponsor immediately
- Taking all appropriate measures requested by the sponsor to resolve the problems found during the audit or inspection

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Documents subject to audit or inspection include but are not limited to all source documents, CRFs, medical records, correspondence, ICFs, EC files, documentation of certification and quality control of supporting laboratories, and records relevant to the study maintained in any supporting pharmacy facilities. Conditions of study material storage are also subject to inspection. In addition, representatives of the sponsor may observe the conduct of any aspect of the clinical study or its supporting activities both within and outside of the investigator's institution.

In all instances, the confidentiality of the data must be respected.

12.4. Study Documentation

12.4.1. Certification of Accuracy of Data

A declaration assuring the accuracy and content of the data recorded on the CRF/eCRF must be signed electronically by the investigator. This signed declaration accompanies each set of participant final CRF/eCRF that will be provided to the sponsor.

12.4.2. Retention of Records

The investigator must retain all essential study documents, including ICFs, source documents, investigator copies of CRFs, and drug accountability records for at least 25 years following the completion or discontinuation of the study, or longer, if a longer period is required by relevant regulatory authorities. The investigator must obtain written approval from the sponsor before discarding or destroying any essential study documents during the retention period following study completion or discontinuation. Records must be destroyed in a manner that ensures confidentiality.

If the investigator's personal situation is such that archiving can no longer be ensured, the investigator must inform the sponsor (written notification) and the relevant records will be transferred to a mutually agreed-upon destination.

12.4.3. Recruitment Strategy

Potential study participants may be identified by the investigator through a variety of means, such as publicizing the trial or using existing participant lists. Recruitment resources may include, but are not limited to, recruitment flyers, brochures, social media ads, newspaper ads, radio ads, etc. A third-party vendor may assist in recruitment efforts, such as the development of recruitment materials. All participant-facing recruitment material, including media advertising and receptionist scripts, will be reviewed and approved by the appropriate EC authority prior to use. Resources may be in paper or electronic form. Regeneron will not access any participant identifiable information as part of recruitment efforts.

12.4.4. Data Protection

Participants will be assigned a unique identifier by the sponsor. Any participant records or datasets that are transferred to the sponsor will contain the identifier only; participant names or any information which would make the participant identifiable will not be transferred.

The participant must be informed that their personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure must also be explained to the

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participant who will be required to give consent for their data to be used as described in the informed consent.

The participant must be informed that their medical records may be examined by Clinical Quality Assurance auditors or other authorized personnel appointed by the sponsor, by appropriate IEC members, and by inspectors from regulatory authorities.

The contract between sponsor and study sites specifies responsibilities of the parties' related data protection, including handling of data security breaches and respective communication and cooperation of the parties.

Information technology systems used to collect, process, and store study-related data are secured by technical and organizational security measures designed to protect such data against accidental or unlawful loss, alteration, or unauthorized disclosure or access.

13. ETHICAL AND REGULATORY CONSIDERATIONS

13.1. Good Clinical Practice Statement

It is the responsibility of both the sponsor and the investigator(s) to ensure that this clinical study will be conducted in accordance with the ethical principles that have their origin in the Declaration of Helsinki, and that are consistent with the ICH guidelines for GCP and applicable regulatory requirements.

13.2. Informed Consent

The principles of informed consent are described in ICH guidelines for GCP.

The ICF used by the investigator must be reviewed and approved by the sponsor prior to submission to the appropriate EC. A copy of the EC-approved ICF and documentation of approval must be provided to the sponsor before study drug will be shipped to the study site.

It is the responsibility of the investigator or designee (if acceptable by local regulations) to obtain written informed consent from each participant prior to study participation and after the aims, methods, objectives, and potential hazards of the study have been explained to the participant in language that the participant can understand. The ICF should be signed and dated by the participant and by the investigator or authorized designee who reviewed the ICF with the participant.

- Participants who can write but cannot read will have the ICF read to them before signing and dating the ICF.
- Participants who can understand but who can neither write nor read will have the ICF read to them in presence of an impartial witness, who will sign and date the ICF to confirm that informed consent was given.

The original ICF must be retained by the investigator as part of the participant's study record, and a copy of the signed ICF must be given to the participant.

If new safety information results in significant changes in the risk/benefit assessment, or if there are significant changes to the study procedures, the ICF must be reviewed and updated appropriately. All study participants must be informed of the new information and provide their

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written consent if they wish to continue in the study. The original signed revised ICF must be maintained in the participant's study record and a copy must be given to the participant.

13.3. Participant Confidentiality and Data Protection

The investigator must take all appropriate measures to ensure that the anonymity of each study participant will be maintained. Participants should be identified by a participant identification number only, on CRFs or other documents submitted to the sponsor. Documents that will not be submitted to the sponsor (eg, signed ICF) must be kept in strict confidence.

The participant's and investigator's personal data, which may be included in the sponsor database, will be treated in compliance with all applicable laws and regulations. The sponsor shall take all appropriate measures to safeguard and prevent access to this data by any unauthorized third party.

13.4. Ethics Committee

An appropriately constituted EC, as described in ICH guidelines for GCP, must review and approve:

- The protocol, ICF, and any other materials to be provided to the participants (eg, advertising) before any participant may be enrolled in the study
- Any amendment or modification to the study protocol or ICF before implementation, unless the change is necessary to eliminate an immediate hazard to the participant, in which case the EC should be informed as soon as possible
- Ongoing studies on an annual basis or at intervals appropriate to the degree of risk

In addition, the EC should be informed of any event likely to affect the safety of participants or the continued conduct of the clinical study.

A copy of the EC approval letter with a current list of the EC members and their functions must be received by the sponsor prior to shipment of drug supplies to the investigator. The approval letter should include the study number and title, the documents reviewed, and the date of the review.

Records of the EC review and approval of all study documents (including approval of ongoing studies) must be kept on file by the investigator.

13.5. Clinical Study Data Transparency

Final study results will be published on public clinical trial registries according to applicable local guidelines and regulations. For data integrity, scientific, and statistical reasons, published results from all participants will be disclosed following End of Study (Section 6.1.4).

Treatment codes will be disseminated to each investigation site thereafter.

For purposes of data disclosure, interim analyses will be available for disclosure when required by local regulations. If the integrity of the ongoing study cannot be ensured or study blinding cannot be maintained, then only final results will be disclosed.

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14. PROTOCOL AMENDMENTS

The sponsor may not implement a change in the design of the protocol or ICF without an EC-approved amendment. Where required per local legislation, regulatory authority approval will also be sought.

15. PREMATURE TERMINATION OF THE STUDY OR CLOSE-OUT OF A SITE

15.1. Premature Termination of the Study

The sponsor has the right to terminate the study prematurely. Reasons may include efficacy, safety, or futility, among others. Should the sponsor decide to terminate the study, the investigator(s) will be notified in writing.

15.2. Close-Out of a Site

The sponsor and the investigator have the right to close-out a site prematurely.

Investigator's Decision

The investigator must notify the sponsor of a desire to close-out a site in writing, providing at least 30 days' notice. The final decision should be made through mutual agreement with the sponsor. Both parties will arrange the close-out procedures after review and consultation.

Sponsor's Decision

The sponsor will notify the investigator(s) of a decision to close-out a study site in writing. Reasons may include the following, among others:

- The investigator has received all items and information necessary to perform the study, but has not enrolled any participant within a reasonable period
- The investigator has violated any fundamental obligation in the study agreement, including but not limited to, breach of this protocol (and any applicable amendments), breach of the applicable laws and regulations, or breach of any applicable ICH guidelines
- The total number of participants required for the study are enrolled earlier than expected

In all cases, the appropriate EC and Health Authorities must be informed according to applicable regulatory requirements, and adequate consideration must be given to the protection of the participants' interests.

16. CONFIDENTIALITY

Confidentiality of information is provided as a separate agreement.

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17. FINANCING AND INSURANCE

Financing and insurance information is provided as a separate agreement.

18. PUBLICATION POLICY

Publication rights and procedures will be outlined in a separate clinical study agreement.

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20. INVESTIGATOR'S AGREEMENT

I have read the attached protocol: **Phase 2 Study of Linvoseltamab in Patients with Smoldering Multiple Myeloma at High Risk of Progression to Multiple Myeloma** and agree to abide by all provisions set forth therein.

I agree to comply with the current International Council for Harmonisation Guideline for Good Clinical Practice and the laws, rules, regulations, and guidelines of the community, country, state, or locality relating to the conduct of the clinical study.

I also agree that persons debarred from conducting or working on clinical studies by any court or regulatory agency will not be allowed to conduct or work on studies for the sponsor or a partnership in which the sponsor is involved. I will immediately disclose it in writing to the sponsor if any person who is involved in the study is debarred, or if any proceeding for debarment is pending, or, to the best of my knowledge, threatened.

This document contains confidential information of the sponsor, which must not be disclosed to anyone other than the recipient study staff and members of the EC. I agree to ensure that this information will not be used for any purpose other than the evaluation or conduct of the clinical investigation without the prior written consent of the sponsor.

_____ _____

(Signature of Investigator) (Date)

_____

(Printed Name)

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APPENDIX 1. INTERNATIONAL MYELOMA WORKING GROUP RESPONSE CRITERIA

Category	Criteria
sCR	• CR as defined below, **plus** • Normal FLC ratio, **and** • Absence of clonal plasma cells in bone marrow by immunohistochemistry or 2- to 4-color flow cytometry[a]
CR[b]	• Negative immunofixation of the serum and urine[c], **and** • Disappearance of any soft tissue plasmacytomas, **and** • <5% plasma cells in bone marrow
VGPR[b]	• Serum and urine M-protein detectable by immunofixation but not on electrophoresis, **or** • ≥90% reduction in serum M-protein plus urine M-protein level <100 mg/24 hours • In addition to the above criteria, if present at baseline, a ≥90% reduction in size of soft tissue plasmacytomas compared with baseline
PR	• ≥50% reduction of serum M-protein plus reduction in urine M-protein by ≥90% or to <200 mg/24 hours • If serum and urine M-protein are not measurable, a ≥50% decrease in the difference between involved and uninvolved FLC levels is required in place of the M-protein criteria • If serum and urine M-protein are not measurable, and serum free light assay is also not measurable, ≥50% reduction in plasma cells is required in place of M-protein, provided baseline bone marrow plasma cell percentage was ≥30% • In addition to the above criteria, if present at baseline, a ≥50% reduction in the size of soft tissue plasmacytomas is also required
MR	• ≥25% to ≤49% reduction of serum M-protein **and** reduction in 24-hour urine M-protein by 50 to 89% • In addition to the above criteria, if present at baseline, a ≥25% to ≤49% reduction in the size of soft tissue plasmacytomas is also required
SD	• Not meeting criteria for CR, VGPR, PR, or PD
PD	• Increase of 25% from lowest confirmed response value in one or more of the following: – Serum M-protein (absolute increase must be ≥0.5 g/dL)[d] – Urine M-protein (absolute increase must be ≥200 mg/24 hours) – Only in patients without measurable serum and urine M-protein levels, the difference between involved and uninvolved FLC levels (absolute increase must be >10 mg/dL) – Only in patients without measurable serum and urine M-protein levels and without measurable involved FLC levels, bone marrow plasma cell percentage (absolute increase must be ≥10%) • Appearance of new osteolytic bone lesions or soft tissue plasmacytomas

sCR=stringent complete response; CR=complete response; VGPR=very good partial response; PR=partial response; MR=minimal response; SD=stable disease; PD=progressive disease; FLC=free light chain; SPD=sum of the products of the maximal perpendicular diameters of measured lesions

[a] Presence or absence of clonal plasma cells is based upon the kappa/lambda ratio. An abnormal kappa/lambda ratio by immunohistochemistry or flow cytometry requires a minimum of 100 plasma cells for analysis. An abnormal ratio reflecting the presence of an aberrant clone is a kappa/lambda ratio of >4:1 or <1:2.

[b] Clarifications to IMWG criteria for coding CR and VGPR in subjects in whom the only measurable disease is by serum FLC levels: CR in such subjects indicates a normal FLC ratio of 0.26 to 1.65 in addition to CR criteria listed above. VGPR in such subjects requires a ≥90% decrease in the difference between involved and uninvolved FLC levels.

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[c] No evidence of residual myeloma-associated monoclonal protein isotype(s) on immunofixation of the serum and urine. New positive immunofixation results after achieving CR, when the isotype is different, can be secondary to therapeutic monoclonal antibodies or monoclonal/oligoclonal immune reconstitution and should be carefully reviewed to prevent confusion with relapse.
[d] M-protein increase of ≥1 g/dL is sufficient to define progression if baseline M-protein is ≥5 g/dL.

Notes:

- All response categories require two consecutive assessments of the applicable disease parameter (serum M-protein, urine M-protein, or serum FLC), performed at any time (no minimum interval is required, it can be done the same day); however, to confirm response or progressive disease, two discrete samples are required; testing cannot be based upon the splitting of a single sample. Radiographic and bone marrow assessments need not be confirmed.
- VGPR and CR categories require serum and urine studies regardless of whether disease at baseline was measurable on serum, urine, both, or neither.
- Patients will be considered to have progressive disease if they meet the criteria for progression by a variable that was not considered measurable at baseline; however, for patients who had a measurable serum or urine M-spike at baseline, progression cannot be defined by increases in serum FLC alone.
- Serum FLC levels should only be used for response assessment when both the serum and urine M-component levels are deemed not measurable at baseline.
- For IgA myeloma, quantitative immunoglobulin measurements are preferred for disease assessments; the same percentage changes apply as for serum M-spike.
- The lowest confirmed value before suspected progression will be used as baseline for calculation of progression; if a serum and/or urine spike is considered too low to quantitate, this value can be assigned as zero as a baseline for documentation of subsequent progressive disease.

Source: Adapted from (Kumar, 2016) and (Rajkumar, 2011)

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APPENDIX 2. CHANGE FROM BASELINE IN SCALE SCORES FOR EORTC QLQ-C30 AND EORTC QLQ-MY20

Change from Baseline and Visit Responses for EORTC QLQ-C30

EORTC QLQ-C30 Scales	Clinically Meaningful Threshold	Sensitivity Threshold	Visit Response
Symptom scales	≤ -10	≤ -sensitivity threshold	Improved
	(-10, 10)	(-sensitivity threshold, sensitivity threshold)	Stable
	≥ +10	≥ +sensitivity threshold	Worsened
Functional scales and global health status	≥ +10	≥ +sensitivity threshold	Improved
	(-10, 10)	(-sensitivity threshold, sensitivity threshold)	Stable
	≤ -10	≤ -sensitivity threshold	Worsened

EORTC QLQ-C30= European Organization for Research and Treatment of Cancer Quality of Life Questionnaire – Core 30

Change from Baseline and Visit Responses for EORTC QLQ-MY20

EORTC QLQ-MY20 Scales	Clinically Meaningful Threshold	Sensitivity Threshold	Visit Response
DS scale	≤ -10	≤ -sensitivity threshold	Improved
	(-10, 10)	(-sensitivity threshold, sensitivity threshold)	Stable
	≥ +10	≥ +sensitivity threshold	Worsened
SE scale	≥ +10	≤ -sensitivity threshold	Improved
	(-10, 10)	(-sensitivity threshold, sensitivity threshold)	Stable
	≤ -10	≥ +sensitivity threshold	Worsened
FP scale	≤ -9	≤ -sensitivity threshold	Improved
	(-9, 9)	(-sensitivity threshold, sensitivity threshold)	Stable
	≥ +9	≥ +sensitivity threshold	Worsened
BI scale	≥ +13	≤ -sensitivity threshold	Improved
	(-13, 13)	(-sensitivity threshold, sensitivity threshold)	Stable
	≤ -13	≥ +sensitivity threshold	Worsened

BI=Body image; DS=Disease symptoms; EORTC QLQ-MY20=European Organization for Research and Treatment of Cancer Quality of Life Questionnaire – Multiple Myeloma Module 20; FP=Future perspective; SE=Side effects of treatment

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APPENDIX 3. ESTIMATED GLOMERULAR FILTRATION RATE

Modification of Diet in Renal Disease (MDRD) Formula

Using standardized S_{Cr} in **mg/dL**, the glomerular filtration rate (GFR) can be estimated using the following MDRD equation:

$$\text{GFR (mL/min/1.73 m}^2) = 175 \times (S_{Cr})^{-1.154} \times (\text{age})^{-0.203} \times (1.212 \text{ if black}) \times (0.742 \text{ if female})$$

If the serum creatinine is obtained using the International System of Units (SI) (ie, micromol/L), use the following formula to convert SI units to conventional (mg/dL) units:

$$\frac{\text{serum creatinine (micromol/L)}}{88.4} = \text{serum creatinine (mg/dL)}$$

Source: (Levey, 2006)

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APPENDIX 4. IMMUNE EFFECTOR CELL-ASSOCIATED ENCEPHALOPATHY SCORE TO ASSESS FOR IMMUNE EFFECTOR CELL-ASSOCIATED NEUROTOXICITY SYNDROME

This CARTOX-10-based assessment tool, termed the ICE score, grades neurotoxicity by evaluating alterations in speech, orientation, handwriting, and concentration that are highly suggestive of the encephalopathy observed in patients with ICANS (Lee, 2019). The ICE score is derived as follows:

- Orientation: orientation to year, month, city, hospital: 4 points
- Naming: ability to name 3 objects (eg, point to clock, pen, button): 3 points
- Following commands: ability to follow simple commands (eg, "Show me 2 fingers" or "Close your eyes and stick out your tongue"): 1 point
- Writing: ability to write a standard sentence (eg, "Our national bird is the bald eagle"): 1 point
- Attention: ability to count backwards from 100 by 10: 1 point.

Total ICE score corresponds to the following ICANS grading:

- 10, no impairment
- 7 to 9, grade 1 ICANS
- 3 to 6, grade 2 ICANS
- 0 to 2, grade 3 ICANS
- 0 due to patient unarousable and unable to perform ICE assessment, grade 4 ICANS

See Section 8.3.3 for management of ICANS.

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SIGNATURE OF SPONSOR'S RESPONSIBLE OFFICERS

(Medical/Study Director, Regulatory Representative, Clinical Study Lead, and Biostatistician)

To the best of my knowledge, this report accurately describes the planned conduct of the study.

Study Title: Phase 2 Study of Linvoseltamab in Patients with Smoldering Multiple Myeloma at High Risk of Progression to Multiple Myeloma

Protocol Number: R5458-ONC-2256 Original

See appended electronic signature page

Sponsor's Responsible Medical/Study Director

See appended electronic signature page

Sponsor's Responsible Regulatory Liaison

See appended electronic signature page

Sponsor's Responsible Clinical Study Lead

See appended electronic signature page

Sponsor's Responsible Biostatistician

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